

Decomposing a graph into linearly many cycles and edges

An exposition of the proof claimed by Ryan Coffey [5]

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Abstract

Erdős and Gallai conjectured in the 1960s that the edge set of every graph on n vertices can be partitioned into $O(n)$ cycles and edges. Bucić and Montgomery came within an iterated logarithm of this, proving the bound $O(n \log^* n)$: they peel off robust sublinear expanders in about $\log^* n$ rounds, and each round costs $O(n)$. We give a complete, reorganised and simplified account of a proof of the full conjecture claimed by Coffey. The new idea is that an expander found in one round is so much richer than anything that happens two rounds later that it can serve as a reservoir of connecting paths for all subsequent rounds. With these reservoirs each vertex pays a constant only at its first two appearances. The few edges that still cannot be closed into cycles are compressed into graphs with at most $n/4$ vertices in total, and these are decomposed by induction.

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1 Introduction

All graphs in this paper are finite and simple. For a graph G let $f(G)$ be the least number of cycles and single edges whose edge sets partition $E(G)$. In the 1960s Erdős and Gallai conjectured that $f(G) = O(n)$ for every graph G on n vertices; see [7] and Problem #184 of [3]. The bound would be best possible up to the constant: cycles contribute even degrees, so in a graph in which every vertex has odd degree, every vertex is an end of a single edge of the decomposition, and at least $n/2$ single edges are needed.

Repeatedly removing a longest cycle gives $f(G) = O(n \log n)$ (Lemma 2.3). Lovász [17] proved that every graph on n vertices decomposes into at most $n/2$ paths and cycles, and Pyber [19] proved that the edges of every such graph can be covered by $n - 1$ cycles and edges. For the decomposition problem, Conlon, Fox and Sudakov [6] proved $f(G) = O(n \log \log n)$, and Bucić and Montgomery [4] proved $f(G) = O(n \log^* n)$, where $\log^*$ is the iterated logarithm. This paper presents a proof of the conjecture.

Theorem 1.1. *There is an absolute constant C such that $f(G) \leq Cn$ for every graph G on n vertices.*

Provenance and status. The argument below is a reorganisation and simplification of a proof claimed by Coffey [5], which in turn builds on the method of Bucić and Montgomery [4]. Coffey reports that his proof was developed with substantial help from an AI system and that it comes with a formalisation in Lean. That formalisation concerns his original manuscript and not the version given here. This exposition was itself prepared with the help of AI systems. It has been checked carefully, but it has not been refereed or formally verified. Section 7 lists the source of every ingredient.

1.1 The strategy

This subsection explains the proof informally. Nothing in it is used later.

Rounds. The method of Bucić and Montgomery [4] runs in rounds. A round starts from a graph $G_r \subseteq G$, on all n vertices of G , with average degree $d = d_r$; write $\lambda = \log d$. First one removes cycles of length at least $d\lambda^2$, one at a time; there are at most $n/(2\lambda^2)$ of them. The graph that remains has no long cycles, so all its expanding subgraphs are small, since expanders contain long cycles. One then splits it into robust sublinear expanders, called *pieces*, of order at most d^2 , deleting a few edges along the way. The deleted edges, together with the edges of a few tiny pieces, form the graph G_{r+1} of the next round, whose average degree is only polylogarithmic in d . After about $\log^* n$ rounds the remaining graph has bounded average degree, and its edges are simply taken as single edges.

In [4] the edges inside the pieces of each round are decomposed using the connectivity of expanders, at a cost proportional to the total order of the pieces, which is $\Theta(n)$ per round. This is where the factor $\log^* n$ comes from: a vertex can lie in pieces of every round, and it is paid for every time.

Anchors. The key observation, due to Coffey [5], is that earlier pieces are vastly richer than anything that happens two rounds later. A piece of round r is an s_r -expander with $s_r = \lambda_r^{100}$, so all its degrees exceed λ_r^{100} . A piece of round $\ell \geq r + 2$ has at most $M_\ell = d_\ell^2$ vertices, and two rounds of polylogarithmic decay give $M_\ell \leq (\log \lambda_r)^{O(1)}$ (Lemma 3.5). So a

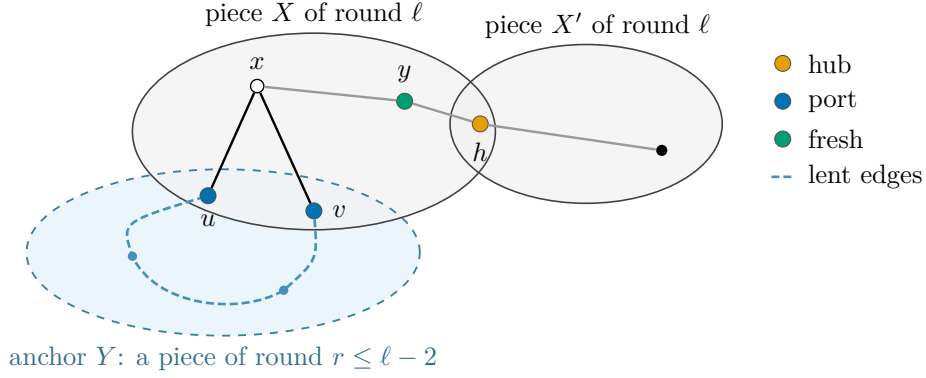


Figure 1.1: Roles of vertices in round ℓ . The hub h lies in two pieces of round ℓ . The ports u, v lie only in X , and both have the old piece Y as their anchor. The edges xu, xv of round ℓ leave Y , and together they form a cherry uxv with both ends in Y . A path of edges lent by Y (dashed) closes the cherry into a cycle. The vertex y is fresh.

piece of round r can set aside, for each later round, a random slice of its edges which is still a robust expander with degrees far larger than anything in that round. One round back is not enough, because a piece of round $r + 1$ may have up to λ_r^{210} vertices.

Fix a round ℓ . A vertex of a piece of round ℓ plays one of three roles (Figure 1.1):

- a *hub* lies in at least two pieces of round ℓ . Hubs are rare: summed over all rounds, the number of incidences between hubs and pieces is $o(n)$;
- a *port* lies in exactly one piece of round ℓ , and also lies in a piece of some round $r \leq \ell - 2$, which we fix and call its *anchor*;
- a *fresh* vertex lies in exactly one piece of round ℓ and has no anchor. Two rounds after a vertex first appears in a piece, it has an anchor, so every vertex is fresh in at most two rounds.

Hubs and fresh vertices are dealt with inside their own piece at a cost of $O(1)$ each, which is $O(n)$ in total. The main work is to deal with ports at a total cost of $o(n)$ over all rounds.

Cherries. An edge of round ℓ never has both ends in an earlier piece, because such edges were assigned to that piece when it was formed. So if u is a port with anchor Y , every edge ux of round ℓ leaves Y . Fix Y and a vertex $x \notin Y$, and pair up the edges from x to ports anchored at Y . This gives *cherries* uxv with both ends in Y . The edges lent by Y can join the last vertex of one cherry to the first vertex of the next, and so chain many vertex-disjoint cherries into a single long cycle (Figure 5.2). In this way almost all edges at ports turn into few cycles. What remains is at most one edge between each vertex x and the ports of each anchor.

The quotient. A typical leftover edge joins a hub h to a port u with anchor Y . We extend it by a private edge uw lent by Y to a vertex w of a sparse random *pool*, and regard the path huw as a single edge hw of an auxiliary graph Q_ℓ whose vertices are hubs and pool vertices. Leftover edges between two ports are treated in the same way, with a pool vertex at each end. A cycle of Q_ℓ lifts to a cycle of G , and a single edge of Q_ℓ lifts to at most two edges of G (Figure 5.3). Since hubs are rare and pools are sparse, the graphs Q_ℓ have at most $n/4$ vertices in total, so we can decompose them by induction on the number of vertices.

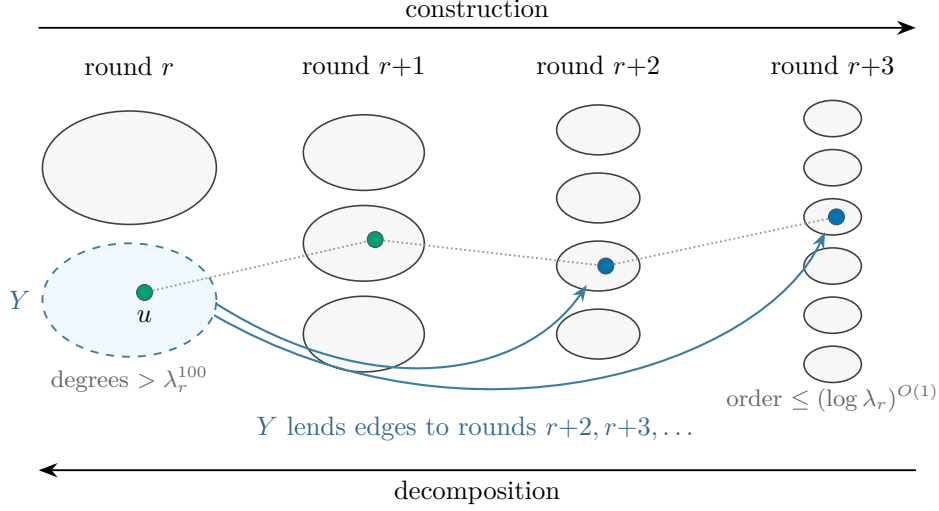


Figure 1.2: The rounds. Pieces get smaller from round to round. A vertex u is fresh (green) in the first two rounds in which it lies in a piece. After that it is a port (blue) whose anchor Y comes from at least two rounds earlier. Every piece lends small random parts of its edges to all rounds at least two steps later. The rounds are constructed from left to right and decomposed from right to left, so every lender is decomposed after all of its borrowers.

Clearing. Each piece is itself a robust expander, so it can absorb all edges among any chosen set P of its vertices into $O(|P|)$ cycles and edges (Lemma 5.1). We use this for the hubs and fresh vertices, and for a few ports that we sacrifice for technical reasons.

Building forwards, decomposing backwards. All rounds are constructed first. The decomposition then goes through the rounds in reverse order (Figure 1.2). So when a piece Y is finally decomposed, every round that borrowed from Y is finished. Only the edges that were actually borrowed are removed from Y . All other edges of Y , including unused lent edges, are decomposed together with its own edges.

The argument thus reduces the problem to graphs with a quarter as many vertices. The precise statement is as follows.

Theorem 1.2 (Reduction). *There is an absolute constant C_0 such that for every graph G on n vertices there are graphs $Q_1, \dots, Q_k$ with*

$$\sum_i |V(Q_i)| \leq n/4 \quad \text{and} \quad f(G) \leq C_0 n + 2 \sum_i f(Q_i).$$

Proof of Theorem 1.1 from Theorem 1.2. We show $f(G) \leq 2C_0 n$ by induction on n . The case $n = 0$ is trivial. For $n \geq 1$ each Q_i has at most $n/4 < n$ vertices, so by induction $f(G) \leq C_0 n + 2 \sum_i 2C_0 |V(Q_i)| \leq C_0 n + C_0 n$. $\square$

Organisation. Section 2 fixes notation and collects the classical tools. Section 3 builds the rounds and proves their structural properties. Section 4 proves the linking lemma, which is the engine behind all the connections we make through expanders. Section 5 proves three self-contained decomposition lemmas: clearing a set of vertices, closing cherries, and building the quotient. Section 6 puts everything together and proves Theorem 1.2. Section 7 lists the sources.

2 Preliminaries

2.1 Notation

For a graph H we write $|H|$ for its number of vertices and $e(H)$ for its number of edges. For $U \subseteq V(H)$, $H[U]$ is the induced subgraph and

$$N_H(U) = \{v \in V(H) \setminus U : v \text{ has a neighbour in } U\}$$

is the *external* neighbourhood of U . For disjoint $A, B \subseteq V(H)$, $E_H(A, B)$ is the set of edges with one end in A and the other in B . For a set F of edges, $H - F$ is obtained by deleting F and keeping all vertices, and $\deg_F(v)$ is the number of edges of F at v . For a set E of edges and a set P of vertices, $E[P]$ is the set of edges of E with both ends in P . All logarithms are to base two; $\ln$ is the natural logarithm.

An *object* is a cycle or a single edge, identified with its edge set. A *decomposition* of a set F of edges is a partition of F into objects, and $f(F)$ is the least number of objects in a decomposition of F ; for a graph H , $f(H) = f(E(H))$. Clearly $f(F) \leq |F|$, and $f(F \cup F') \leq f(F) + f(F')$ for disjoint F, F' . To *pay for* an edge means to use it as a single-edge object.

A path is *through* a set V if all its interior vertices lie in V . Its ends are unrestricted, and a single edge is a path through every set.

Definition 2.1 (Linking). Let R be a graph, $V \subseteq V(R)$ and $t \geq 1$. We say that R *links through* V *with multiplicity* t if the following holds for every multiset $\mathcal{P}$ of pairs of distinct vertices of R in which every vertex lies in at most t pairs, counted with multiplicity: there are pairwise edge-disjoint paths P_π , $\pi \in \mathcal{P}$, in R , such that P_π joins the two vertices of π and is through V .

This is the notion of path connectivity of [4, Definition 7], without the bound on the lengths of the paths, which we never need.

2.2 Three classical facts

Lemma 2.2 (Paths; Lovász [17], see [4, Corollary 22]). *The edges of every graph can be decomposed into paths so that every vertex is an end of at most two of the paths. In particular, if every edge of a graph has both ends in a set W , then its edges decompose into at most $|W|$ paths.*

The second statement follows from the first because every path has two distinct ends.

Lemma 2.3 (Long cycles). *Every graph H on $h \geq 1$ vertices satisfies $f(H) \leq h(\log h + 1)$.*

Proof. If a graph on h vertices has $m > h$ edges, then it contains a cycle of length greater than m/h . Indeed, delete vertices of degree less than m/h one at a time. Each deletion removes fewer than m/h edges and at most h vertices are deleted, so fewer than m edges are removed, and the process stops at a nonempty subgraph of minimum degree at least $m/h > 1$. All neighbours of the first vertex v_0 of a longest path $v_0 v_1 \cdots v_k$ in this subgraph lie on the path. If v_i is the last of them, then $i \geq m/h$, and $v_0 v_1 \cdots v_i v_0$ is a cycle of length $i + 1 > m/h$.

Now remove such cycles from H while it has more than h edges. While the number m of remaining edges lies in $(2^i h, 2^{i+1} h]$, every removed cycle has more than 2^i edges, so at most h cycles are removed. As $m \leq h^2/2$, only the ranges with $2^i < h/2$ occur, and there are at most $\log h$ of them. So at most $h \log h$ cycles are removed, and at most h edges remain, which we pay for. $\square$

This is the classical bound $f(G) = O(n \log n)$, observed already by Erdős and Gallai; see [6, 4].

Theorem 2.4 (Haxell [11]). *Let $h \geq 1$ and let $\mathcal{A}_1, \dots, \mathcal{A}_q$ be families of sets of size at most h . Suppose that for every nonempty $I \subseteq [q]$ and every set S with $|S| \leq (2h - 1)(|I| - 1)$, some member of some $\mathcal{A}_i$ with $i \in I$ is disjoint from S . Then there are pairwise disjoint sets $A_1 \in \mathcal{A}_1, \dots, A_q \in \mathcal{A}_q$.*

2.3 Probability

Lemma 2.5 (Chernoff bounds; see [12, Theorems 2.1 and 2.8]). *Let X be a sum of independent indicator variables and $0 < \delta \leq 1$. If $\mathbb{E}X \geq \mu$ then $\mathbb{P}(X \leq (1 - \delta)\mu) \leq e^{-\delta^2\mu/2}$, and if $\mathbb{E}X \leq \mu$ then $\mathbb{P}(X \geq (1 + \delta)\mu) \leq e^{-\delta^2\mu/3}$.*

We also need a Bernstein-type inequality for functions of independent bits. It follows by applying Freedman's martingale inequality [8] to the exposure martingale; see also [18].

Lemma 2.6 (Bernstein inequality for bounded differences). *Let $\xi_1, \dots, \xi_k$ be independent $\{0, 1\}$ -valued random variables with $\mathbb{P}(\xi_i = 1) = p_i$, and let $Z = \phi(\xi_1, \dots, \xi_k)$, where changing ξ_i alone changes ϕ by at most $c_i \leq c$. Put $\beta = \sum_i p_i(1 - p_i)c_i^2$. Then for all $\lambda > 0$,*

$$\mathbb{P}(Z \leq \mathbb{E}Z - \lambda) \leq \exp\left(-\frac{\lambda^2}{2(\beta + c\lambda/3)}\right).$$

Proof sketch. Let $Z_i = \mathbb{E}[Z \mid \xi_1, \dots, \xi_i]$. Given $\xi_1, \dots, \xi_{i-1}$, let $g(x) = \mathbb{E}[Z \mid \xi_1, \dots, \xi_{i-1}, \xi_i = x]$. By independence $g(1) - g(0)$ is an average of differences of ϕ , so $|g(1) - g(0)| \leq c_i$. Hence the martingale difference $Z_i - Z_{i-1} = g(\xi_i) - \mathbb{E}g(\xi_i)$ is bounded by c in absolute value and has conditional variance $p_i(1 - p_i)(g(1) - g(0))^2 \leq p_i(1 - p_i)c_i^2$. Freedman's inequality [8] gives the bound. $\square$

2.4 Robust sublinear expanders

Throughout the paper we fix

$$\varepsilon = 1/32.$$

Definition 2.7 (Expanders). Let $s \geq 0$. A graph H on $N \geq 2$ vertices is an s -expander if

$$|N_{H-F}(U)| \geq \frac{\varepsilon|U|}{\log^2 N} \quad \text{whenever } 1 \leq |U| \leq \frac{2}{3}N \text{ and } F \subseteq E(H), |F| \leq s|U|.$$

Sublinear expanders were introduced by Komlós and Szemerédi [13, 14]. The robust form, in which up to $s|U|$ edges may be deleted, was introduced independently by Haslegrave, Kim and Liu [10] and by Sudakov and Tomon [20]. In the notation of [4, Definition 11], our s -expanders are the (ε, s) -expanders with $\varepsilon = 2^{-5}$. See [16] for a survey. We use two simple consequences of the definition.

- An s -expander has minimum degree greater than s . Indeed, for a vertex v , taking $U = \{v\}$ and F the set of edges at v gives $N_{H-F}(U) = \emptyset$, so $|F| > s$. In particular $N > s + 1$.
- An s -expander is an s' -expander for every $s' \leq s$.

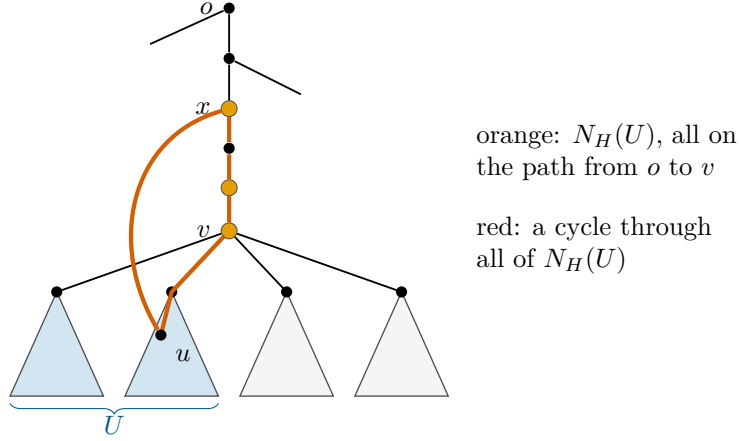


Figure 3.1: Proof of Lemma 3.1. In a depth-first search tree there are no edges between different subtrees of v . So the external neighbourhood of the shaded union U of subtrees lies on the path from the root o to v , and the highest neighbour x closes a cycle through all of it.

3 Expanders and the hierarchy of rounds

3.1 Long cycles

The rounds work because expanders contain long cycles. Bucić and Montgomery use a depth-first search to find a cycle of length $\Omega(N/\log^4 N)$ in a 0-expander on N vertices [4, Lemma 25]. A slightly different use of depth-first search, a now standard tool for long paths and cycles [15], gives a better bound with less work.

Lemma 3.1 (Long cycles in expanders). *Every 0-expander H on $N \geq 2^{20}$ vertices contains a cycle of length at least $N/(2^8 \log^2 N)$.*

Proof. H is connected: otherwise a smallest component K has $|K| \leq N/2$ and $N_H(K) = \emptyset$, which contradicts expansion. Let T be a depth-first search tree of H with root o . Every edge of H joins a vertex to one of its ancestors in T . For a vertex y let T_y be the set of its descendants, including y .

Put $k = \lfloor N/2 \rfloor$, and let v be a vertex of maximum depth with $|T_v| > k$. The sets T_c , for the children c of v , partition $T_v \setminus \{v\}$. Each has size at most k , and together they have size at least k . So we can choose a union U of some of them with $k/2 \leq |U| \leq k$: take one T_c of size at least $k/2$ if there is one, and otherwise add sets T_c one at a time until the size first reaches $k/2$.

Since U is a union of whole subtrees hanging from v , every vertex outside U with a neighbour in U is an ancestor of that neighbour, and hence lies on the path from o to v in T (Figure 3.1). Note that v itself lies in $N_H(U)$. As $|U| \leq N/2$, expansion gives

$$q := |N_H(U)| \geq \frac{\varepsilon k}{2 \log^2 N} \geq \frac{N}{2^8 \log^2 N} \geq 2.$$

Let x be the vertex of $N_H(U)$ closest to o , and let $u \in U$ be a neighbour of x . As $q \geq 2$, x is a proper ancestor of v . So the path from x to u in T has length at least two, and it passes through all q vertices of $N_H(U)$. Together with the edge ux it forms a cycle of length at least $q + 1$. $\square$

3.2 Splitting into expanders

A graph that is not an s -expander has a set U with a small external neighbourhood once at most $s|U|$ edges are deleted. Such a set splits the graph into two smaller graphs that overlap only in a small separator. Repeating this gives a decomposition into expanders. This is the method of Bucić and Montgomery [4, Lemma 14]. Coffey [5] added a rule that moves vertices with many deleted edges into the separator. This makes the deleted edges a thin cut, which is property (iii) below.

Lemma 3.2 (A splitting step). *Let $s \geq 0$ and $\tau > 0$, and let H be a graph on $m \geq 2$ vertices which is not an s -expander, with $\tau \geq 256s \log^2 m$. Then there is a partition $V(H) = A \sqcup B \sqcup C$ with $A \neq \emptyset$ such that*

- (i) $|B| \leq 2\varepsilon|A|/\log^2 m$ and $|A| + |B| < 3m/4$;
- (ii) the set $F = E_H(A, C)$ of deleted edges satisfies $|F| \leq 2s|A|$, and $F = \emptyset$ if $s = 0$;
- (iii) every vertex of A has fewer than τ neighbours in C , and every vertex of C has fewer than τ neighbours in A .

The two children of H are

$$H_1 = H[A \cup B] \quad \text{and} \quad H_2 = H[B \cup C] - E(H[B]).$$

They meet in the separator B , both have fewer than m vertices, and $E(H)$ is the disjoint union of $E(H_1)$, $E(H_2)$ and F (Figure 3.2).

Proof. Choose U and F' with $1 \leq |U| \leq 2m/3$, $|F'| \leq s|U|$ and $|N_{H-F'}(U)| < \varepsilon|U|/\log^2 m$. Let $N = N_{H-F'}(U)$ and $C_0 = V(H) \setminus (U \cup N)$. All edges between U and C_0 lie in F' , so $F_0 = E_H(U, C_0)$ has at most $s|U|$ edges. Let U^+ be the set of vertices $u \in U$ with $\deg_{F_0}(u) \geq \tau$, and let $A = U \setminus U^+$. Let C^+ be the set of vertices of C_0 with at least τ neighbours in A . Put

$$B = N \cup U^+ \cup C^+, \quad C = C_0 \setminus C^+.$$

The edges of F_0 at vertices of U^+ and the edges between C^+ and A are disjoint subsets of F_0 . Hence $\tau(|U^+| + |C^+|) \leq s|U|$, and so $|U^+| + |C^+| \leq |U|/(256 \log^2 m) = \varepsilon|U|/(8 \log^2 m)$. Thus $|A| \geq \frac{255}{256}|U| > 0$ and

$$|B| < \frac{9\varepsilon|U|}{8 \log^2 m} \leq \frac{2\varepsilon|A|}{\log^2 m}, \quad |A| + |B| \leq |U| + |B| \leq \frac{2}{3}m(1 + \frac{9}{256}) < \frac{3}{4}m.$$

Every edge between $A \subseteq U$ and $C \subseteq C_0$ lies in F_0 , so $|F| \leq s|U| \leq 2s|A|$. If $s = 0$ then $F_0 = \emptyset$. Finally, a vertex of A has $\deg_{F_0} < \tau$, and a vertex of C has fewer than τ neighbours in A by the definition of C^+ . $\square$

A *splitting tree* for a graph H_0 is a rooted binary tree whose nodes are graphs and whose root is H_0 . Each node H that is not a leaf is split by Lemma 3.2, for some parameters s_H, τ_H such that H is not an s_H -expander and $\tau_H \geq 256s_H \log^2 |H|$. Its two children are the two children of the split. The edge set of H_0 is the disjoint union of the edge sets of the leaves and the sets of deleted edges at all splits.

Every vertex of a node lies in some leaf below it, because the children of a node cover its vertex set. A vertex lies in two or more leaves if and only if it lies in the separator of some split. Indeed, a vertex of a separator lies in both children, and a vertex lying in two leaves lies in both children of their lowest common ancestor. Such vertices are *shared*, and the others are *private*. For a leaf Y we write $\text{sh}(Y)$ for its set of shared vertices. A private vertex of Y lies in exactly the nodes on the path from the root to Y .

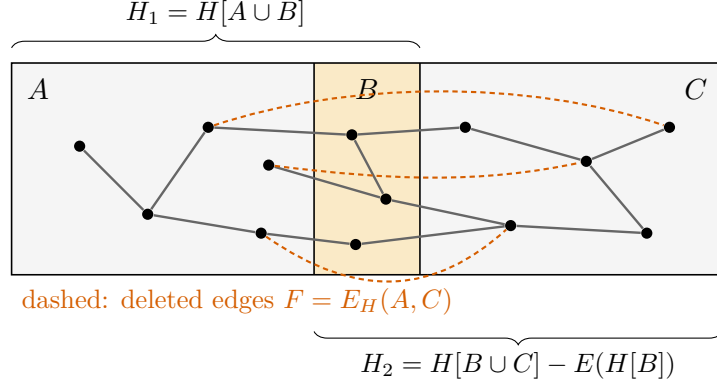


Figure 3.2: A splitting step (Lemma 3.2). The separator $B = N \cup U^+ \cup C^+$ consists of the small boundary N of $U = A \cup U^+$ and the vertices that would otherwise have many deleted edges. The children overlap only in B ; edges inside B go to H_1 . The few deleted edges between A and C form a thin cut: no vertex has τ or more of them.

Lemma 3.3 (Splitting trees). *Let $\mathcal{T}$ be a splitting tree for a graph H_0 on n_0 vertices.*

- (a) *The leaves of $\mathcal{T}$ have total order less than $4n_0/3$.*
- (b) *For every $T \geq 2$, $\sum_Y |\text{sh}(Y)| < n_0/\log T$, where the sum is over the leaves Y with $|Y| \geq T$.*
- (c) *If $s_H \leq s$ for every split, then at most $7sn_0 \log n_0$ edges are deleted in total.*
- (d) *Suppose that $\tau_H \leq \tau$ at every split that deletes at least one edge. Then for every leaf Y and every vertex h , fewer than τ deleted edges join h to private vertices of Y , and none do if $h \in Y$.*

Proof. For every split of a node H , with parts A, B, C , and every $a \in A$, choose a leaf below the first child that contains a , and call the pair (leaf, a) a *copy*. Consider a fixed copy (Y, v) , and let $H^{(1)} \supsetneq H^{(2)} \supsetneq \dots$ be the nodes whose splits chose it. Each $H^{(i+1)}$ lies below the first child of $H^{(i)}$, so $|H^{(i+1)}| < \frac{3}{4}|H^{(i)}|$. Consequently, if all these nodes have order at least $T \geq 2$, then the j -th smallest has order at least $T(4/3)^{j-1}$.

(a),(b) Let Δ_T be the sum of $|B|$ over the splits of nodes of order at least T . Spread the bound $|B| \leq \sum_{a \in A} 2\varepsilon/\log^2 |H|$ of Lemma 3.2 over the copies chosen at the split. A copy then receives at most

$$2\varepsilon \sum_{j \geq 0} \frac{1}{(\log T + j \log \frac{4}{3})^2} \leq 2\varepsilon \left(\frac{1}{\log^2 T} + \frac{1}{\log \frac{4}{3} \cdot \log T} \right) < \frac{7\varepsilon}{\log T}$$

from splits of order at least T . Replacing a node by its two children increases the total order by $|B|$, so the total order of the leaves is $S = n_0 + \Delta_2$. As there are at most S copies, $\Delta_T < 7\varepsilon S/\log T$. For $T = 2$ this gives $S < n_0 + 7\varepsilon S$, so $S < n_0/(1 - 7\varepsilon) < 4n_0/3$, which proves (a). Hence $\Delta_T < \frac{7}{24}n_0/\log T$.

For (b), let c_v be the number of leaves of order at least T in which v is shared, and k_v the number of splits of nodes of order at least T with v in the separator. We claim $c_v \leq 2k_v$. Suppose $c_v \geq 1$. The leaves counted by c_v , together with their ancestors, form a subtree whose nodes all have order at least T and contain v . If $c_v \geq 2$, this subtree has at least $c_v - 1$ nodes with two children in it, and v lies in the separator of each of them. If $c_v = 1$, then v also lies in another leaf, and v lies in the separator of the lowest common ancestor of the two leaves, which has order at least T . Thus $\sum_Y |\text{sh}(Y)| = \sum_v c_v \leq 2\Delta_T < n_0/\log T$.

(c) Spread the bound $|F| \leq 2s|A|$ over the copies chosen at the split, $2s$ each. The orders of the nodes charging a fixed copy decrease by a factor $3/4$ and lie between 2 and n_0 , so there are at most $1 + \log_{4/3}(n_0/2) \leq 2.41 \log n_0$ of them. Hence at most $2s \cdot \frac{4}{3}n_0 \cdot 2.41 \log n_0 \leq 7sn_0 \log n_0$ edges are deleted.

(d) Let hy be a deleted edge with y private to Y , deleted at the split of a node H . Since y is private, H is an ancestor of Y , and y lies in the part (A or C) of the child towards Y . So h lies in the other part and not in the child towards Y , and therefore $h \notin Y$. Moreover, H is the last node on the path from the root to Y that contains h . So all the deleted edges in question are deleted at the same split. Since h and the private vertices of Y lie in opposite parts of this split, there are fewer than τ of them by [Lemma 3.2\(iii\)](#). $\square$

3.3 The rounds

We fix a large absolute constant D . Every statement from now on holds if D is larger than a suitable absolute constant, and we assume this tacitly. We write $o_D(n)$ for any quantity bounded by $\eta(D)n$, where η is a function that depends only on D and tends to 0 as $D \rightarrow \infty$.

For a real number $d \geq D$ we use the parameters

$$\lambda = \log d, \quad M = \lceil d^2 \rceil, \quad m = \lceil \lambda^{103} \rceil, \quad s = \lceil \lambda^{100} \rceil, \quad \tau = \lceil 2^{14} \lambda^{102} \rceil,$$

with the meanings shown in [Table 1](#). The exponents are not optimised. We only need $\tau \geq 256s \log^2 M$, which holds because $\log M \leq 2\lambda + 1$, and the comfortable gaps between s , τ and m .

symbol	value	role
d_r	$2e(G_r)/n$	average degree at the start of round r
λ_r, μ_r	$\log d_r, \log \lambda_r$	logarithms of the degree
M_r	$\lceil d_r^2 \rceil$	maximum order of a piece of round r
m_r	$\lceil \lambda_r^{103} \rceil$	minimum order of a piece of round r
s_r	$\lceil \lambda_r^{100} \rceil$	pieces of round r are s_r -expanders
τ_r	$\lceil 2^{14} \lambda_r^{102} \rceil$	thin-cut threshold for splits in round r

Table 1: The parameters of round r .

Construction. Let $G_1 = G$. For $r = 1, 2, \dots$ let $d_r = 2e(G_r)/n$. If $d_r < D$ we stop and put $R = r - 1$. Otherwise we carry out round r , with the parameters of [Table 1](#):

- (R1) *Long cycles.* Repeatedly remove from G_r a cycle of length at least $d_r \lambda_r^2$, as long as there is one. Let $\mathcal{C}_r$ be the set of removed cycles and G'_r the remaining graph.
- (R2) *Splitting.* Build a splitting tree $\mathcal{T}_r$ for G'_r in two stages. In the first stage, split every node with at least two vertices that is not a 0-expander, with $s_H = 0$. The leaves of the first stage are 0-expanders or single vertices. In the second stage, split every node of order at least m_r that is not an s_r -expander, with $(s_H, \tau_H) = (s_r, \tau_r)$. The leaves of order at least m_r are s_r -expanders and are called the *pieces* of round r . For a piece Y (a set of vertices) we write H_Y for the leaf itself, an s_r -expander with vertex set Y .
- (R3) *Assignment.* Assign every edge of every H_Y to Y . Then assign every other edge of G'_r with both ends in some piece to one such piece. Let $E_Y \supseteq E(H_Y)$ be the set of edges assigned to Y .
- (R4) *Next round.* G_{r+1} consists of the edges of G'_r that were not assigned, on the vertex set $V(G)$.

We write $\mathcal{Y}_r$ for the set of pieces of round r , and $\text{sh}_r(Y)$ for the shared vertices of $Y \in \mathcal{Y}_r$ in $\mathcal{T}_r$.

Lemma 3.4 (The hierarchy). *For every round $r \leq R$:*

- (H1) *every piece $Y \in \mathcal{Y}_r$ has $m_r \leq |Y| \leq M_r$;*
- (H2) *$\sum_{Y \in \mathcal{Y}_r} |Y| < 4n/3$, so $|\mathcal{Y}_r| < 4n/(3m_r)$; and $\sum_{Y \in \mathcal{Y}_r} |\text{sh}_r(Y)| < n/\log m_r$;*
- (H3) *$|\mathcal{C}_r| \leq n/(2\lambda_r^2)$;*
- (H4) *$d_{r+1} \leq \lambda_r^{105}$;*
- (H5) (separation) *for every $\ell > r$, no edge of G_ℓ has both ends in a piece of round r ;*
- (H6) (thin cuts) *for every $Y \in \mathcal{Y}_r$ and every vertex h , fewer than τ_r edges of G_{r+1} join h to vertices of $Y \setminus \text{sh}_r(Y)$.*

Moreover, $E(G)$ is the disjoint union of the cycles in $\bigcup_r \mathcal{C}_r$, the sets E_Y over all pieces of all rounds, and $E(G_{R+1})$, where $e(G_{R+1}) < Dn/2$. The graphs H_Y over all pieces of all rounds are pairwise edge-disjoint.

Proof. (H1) Let W be a leaf of the first stage, a 0-expander, and suppose $|W| > M_r$. By Lemma 3.1 and since $x/\log^2 x$ is increasing for $x \geq 8$, W contains a cycle of length at least $M_r/(2^8 \log^2 M_r) \geq d_r^2/(2^8 \cdot 9\lambda_r^2) > d_r \lambda_r^2$. But $W \subseteq G'_r$, which has no such cycle. So the first-stage leaves, and hence all pieces, have order at most M_r .

(H2) This follows from Lemma 3.3(a) and (b) with $T = m_r$.

(H3) Each removed cycle has at least $d_r \lambda_r^2$ of the $d_r n/2$ edges of G_r .

(H4) Every edge of G_{r+1} is deleted in $\mathcal{T}_r$ or lies in a leaf of order less than m_r . The first stage deletes nothing. Below each first-stage leaf W the second stage is a splitting tree in which $s_H \leq s_r$. So by Lemma 3.3(c) it deletes at most $7s_r |W| \log M_r$ edges. The first-stage leaves have total order less than $4n/3$, so at most $7s_r \cdot \frac{4}{3}n(2\lambda_r + 1) \leq 40\lambda_r^{101}n$ edges are deleted. A leaf K of order less than m_r has fewer than $|K|m_r/2$ edges, and all leaves have total order less than $4n/3$. Hence $d_{r+1} \leq 80\lambda_r^{101} + \frac{4}{3}m_r \leq \lambda_r^{105}$.

(H5) Every edge of G'_r with both ends in a piece of round r was assigned in (R3), and $G_\ell \subseteq G_{r+1}$ for $\ell > r$.

(H6) Let $hu \in E(G_{r+1})$ with $u \in Y \setminus \text{sh}_r(Y)$. If hu lay in a leaf of $\mathcal{T}_r$, that leaf would contain u , so it would be Y , and hu would have been assigned to Y . So hu is a deleted edge, and Lemma 3.3(d) applies. Only second-stage splits delete edges, and they use τ_r .

The final statements follow from the construction, since $d_{R+1} < D$. $\square$

Lemma 3.5 (Scales). *Let $\mu_r = \log \lambda_r$.*

- (a) *For $r < R$: $\lambda_{r+1} \leq \lambda_r/2$, $\mu_{r+1} \leq \mu_r/2$ and $M_{r+1} \leq M_r/2$. Also $\mu_R \geq \log \log D$ and $M_R \geq D^2$.*
- (b) *For $3 \leq \ell \leq R$ and $r \leq \ell - 2$: $M_\ell \leq \mu_{\ell-2}^{211} \leq \mu_r^{211}$, and $2^{\ell-r} \leq \lambda_r$.*
- (c) *For fixed $a > 0$ and $k \geq 0$, $\sum_{r=1}^R (R-r+1)^k \mu_r^{-a} = O_{a,k}((\log \log D)^{-a})$. The same holds with μ_r replaced by λ_r , m_r or M_r , which are larger.*

Proof. (a) By (H4), $\lambda_{r+1} \leq 105 \log \lambda_r = 105\mu_r \leq \lambda_r/2$ and $\mu_{r+1} \leq \log(105\mu_r) \leq \mu_r/2$, because $\mu_r \geq \log \log D$ is large. Also $d_{r+1} \leq \lambda_r^{105} \leq d_r/2$, which gives $M_{r+1} \leq M_r/2$. (b) We have $M_\ell \leq 2d_\ell^2 \leq 2\lambda_{\ell-1}^{210} \leq 2(105\mu_{\ell-2})^{210} \leq \mu_{\ell-2}^{211}$. By (a), $\lambda_r \geq 2^{\ell-r}\lambda_\ell \geq 2^{\ell-r}$. (c) By (a), $\mu_r \geq 2^{R-r} \log \log D$, so the sum is at most $(\log \log D)^{-a} \sum_{j \geq 0} (j+1)^k 2^{-aj}$. $\square$

Part (c) is the only place where the number of rounds enters. It is why no factor $\log^* n$ appears: quantities charged once per round, or even once per pair of rounds, are summable because the parameters grow like a tower towards earlier rounds.

3.4 Roles

Definition 3.6 (Hubs, residents, ports and anchors). Fix a round $\ell \leq R$. A *hub* of round ℓ is a vertex that lies in at least two pieces of round ℓ . Let Hub_ℓ be the set of hubs. A vertex that lies in exactly one piece X of round ℓ is a *resident* of X . A resident u is a *port* if some piece of a round $r \leq \ell - 2$ contains u . In that case we fix one such piece and call it the *anchor* $\text{anc}_\ell(u)$; this choice depends only on the hierarchy, not on any of the random choices made later. A resident that is not a port is *fresh*, and Fresh_ℓ is the set of fresh residents.

For a piece Y of round r and a round $\ell \geq r + 2$, the *crowding* of Y in round ℓ is

$$\kappa_{Y,\ell} = \max_{h \in V(G)} |\{u : u \text{ is a port of round } \ell \text{ with } \text{anc}_\ell(u) = Y \text{ and } hu \in E(G_\ell)\}|.$$

The number of cycles needed to close the cherries of an anchor grows with its crowding (Lemma 5.2), so we need the crowding to be small on average. Let ν_ℓ be the number of pieces of rounds at most $\ell - 2$.

Lemma 3.7 (Bookkeeping).

- (a) $\sum_\ell |\text{Fresh}_\ell| \leq 2n$.
- (b) $\sum_\ell \sum_{X \in \mathcal{Y}_\ell} |X \cap \text{Hub}_\ell| = o_D(n)$.
- (c) $\kappa_{Y,\ell} < \tau_r + |\text{sh}_r(Y)|$ for $Y \in \mathcal{Y}_r$, and $\sum_\ell \sum_{r \leq \ell-2} \sum_{Y \in \mathcal{Y}_r} \kappa_{Y,\ell} = o_D(n)$.
- (d) $\nu_\ell \leq 3n/m_{\ell-2}$, and $\sum_{\ell=3}^R \nu_\ell M_\ell^a = o_D(n)$ for every fixed $a > 0$.
- (e) $\sum_r |\mathcal{C}_r| = o_D(n)$.

Proof. (a) Let r_0 be the first round in which a vertex u lies in a piece. In every round $\ell \geq r_0 + 2$ in which u is a resident, the piece of round r_0 containing u makes u a port. So u is fresh in at most the two rounds r_0 and $r_0 + 1$.

(b) A hub lies in two leaves of $\mathcal{T}_\ell$, so $X \cap \text{Hub}_\ell \subseteq \text{sh}_\ell(X)$. By (H2) the inner sum is less than $n/\log m_\ell \leq n/(103\mu_\ell)$. Now sum using Lemma 3.5(c).

(c) Let u be a port of round ℓ with anchor $Y \in \mathcal{Y}_r$ and $hu \in E(G_\ell) \subseteq E(G_{r+1})$. If $u \notin \text{sh}_r(Y)$, there are fewer than τ_r such u for a fixed h , by (H6). All other such u lie in $\text{sh}_r(Y)$. Summing over $Y \in \mathcal{Y}_r$ with (H2), and then over the at most $R - r$ rounds ℓ , the total is at most

$$\sum_r (R - r) \left(\frac{4n}{3m_r} \tau_r + \frac{n}{\log m_r} \right) \leq \sum_r (R - r) \left(\frac{2^{15}}{\lambda_r} + \frac{1}{103\mu_r} \right) n = o_D(n).$$

(d) By (H2) and Lemma 3.5(a), $\nu_\ell < \sum_{r \leq \ell-2} 4n/(3m_r) \leq 3n/m_{\ell-2}$. By Lemma 3.5(b), $\nu_\ell M_\ell^a \leq 3n\mu_{\ell-2}^{211a}/\lambda_{\ell-2}^{103} \leq n/\lambda_{\ell-2}$, and the sum is $o_D(n)$ by Lemma 3.5(c).

(e) This follows from (H3) and Lemma 3.5(c). $\square$

4 Linking through sparse random sets

Every cycle we build through an expander is closed by a path through a sparse random set of vertices. This section proves that robust expanders support such paths, for many pairs at once. First we note that a random sample of the edges of an expander is again an expander.

Lemma 4.1 (Sampling edges). *Let H be an s -expander on N vertices, and let $0 < p \leq 1$ with $ps \geq 100 \log N$. Let H_p be the random spanning subgraph of H that contains every edge independently with probability p . Then H_p is a $(ps/2)$ -expander with probability at least $1 - N^{-10}$.*

Proof. Suppose H_p is not a $(ps/2)$ -expander. Then there are U with $1 \leq u = |U| \leq 2N/3$ and $F \subseteq E(H_p)$ with $|F| \leq psu/2$ such that $T = N_{H_p-F}(U)$ has fewer than $\varepsilon u / \log^2 N$ vertices. All edges of H_p between U and $V \setminus (U \cup T)$ lie in F . On the other hand, H has more than su edges between U and $V \setminus (U \cup T)$: otherwise deleting them would leave the external neighbourhood of U inside T , which contradicts expansion. So for this pair (U, T) , at most $psu/2$ of more than su given edges survive in H_p . By Lemma 2.5 this has probability at most $e^{-psu/8} \leq N^{-18u}$. There are at most N^{2u} pairs (U, T) with $|U| = u$, since $|T| < u$. Hence the failure probability is at most $\sum_{u \geq 1} N^{-16u} \leq N^{-10}$. $\square$

The following lemma is the engine of the proof. Bucić and Montgomery proved it for random sets of density $1/3$ and a fixed polylogarithmic multiplicity [4, Theorem 16]. We need sparse random sets and arbitrary multiplicity. We give a complete proof that follows their strategy, with two simplifications: a single bounded-degree certificate (Claim 4.2.3) replaces their two-case analysis [4, Proposition 13], and Haxell's theorem replaces the hypergraph matching theorem of Aharoni and Haxell [1].

Lemma 4.2 (Linking). *There is an absolute constant C_* such that the following holds. Let $t \geq 1$ and $0 < \rho \leq 1/2$, let H be an s -expander on N vertices with $s \geq C_* t \rho^{-3} \log^{18} N$, and let $V \subseteq V(H)$ be random, containing every vertex independently with probability ρ . Then with probability at least $1 - N^{-3}$, H links through V with multiplicity t .*

Linking through V implies linking through every superset of V . Hence the lemma also applies if every vertex is included independently with probability at least ρ : a ρ -random set can be coupled inside such a set.

Proof. Write $L = \log N$ and $\alpha = \varepsilon/L^2$. Since $N > s \geq C_* L^{18}$, the order N is as large as we like if C_* is large. For a graph H' on $V(H)$, a set U , a set F of edges and an integer $k \geq 0$, the ball $B_{H'-F}^k(U)$ is the set of vertices of V that can be reached from U by a path in $H' - F$ through V of length at most k . Paths of length 0 are allowed, so $U \cap V \subseteq B_{H'-F}^k(U)$. We use the parameters

$$k = \left\lceil \frac{100L}{\alpha} \right\rceil, \quad p = \frac{\rho}{10k}, \quad \theta = \frac{C_1 L}{\alpha p}, \quad \mu_0 = \frac{\alpha}{6\theta}, \quad h = \lceil 4kL \rceil, \quad \bar{s} = \frac{40ht}{\rho}, \quad K = \left\lceil \frac{\bar{s}}{\mu_0} \right\rceil,$$

where C_1 is a large absolute constant. Then $k = O(L^3)$, $\theta = O(L^6/\rho)$, $\mu_0^{-1} = O(L^8/\rho)$, $h = O(L^4)$ and $K = O(tL^{12}\rho^{-2})$. So $2K(\theta + 1) = O(tL^{18}\rho^{-3}) \leq s$ if C_* is large. The proof has three parts: balls give paths (Claims 4.2.1–4.2.2), large balls exist in expanders with enough robustness (Claims 4.2.3–4.2.7), and splitting the edges into colours provides that robustness.

From balls to paths. The first claim is [4, Proposition 8].

Claim 4.2.1. *Let H' be a graph on $V(H)$ and $u \geq 1$ with $|V| \geq 4u - 2$, and suppose $|B_{H'}^k(U)| > |V|/2$ for every set U of u vertices. Let $\{x_1, y_1\}, \dots, \{x_{2u-1}, y_{2u-1}\}$ be pairs, all of whose $4u - 2$ vertices are distinct. Then some x_i and y_i are joined by a path in H' through V of length at most $4kL$.*

Proof. Call x good if $|B_{H'}^{2kL}(\{x\})| > |V|/2$. If both x_i and y_i are good, their balls meet in a vertex $z \in V$, and joining the two paths at z gives a walk through V of length at most $4kL$. This walk contains the required path. So it suffices to show that fewer than u vertices are bad.

Suppose S_0 is a set of u bad vertices. We find sets $S_0 \supseteq S_1 \supseteq \dots$ with $|S_{i+1}| \leq \lceil |S_i|/2 \rceil$ and $|B_{H'}^{(i+1)k}(S_i)| > |V|/2$, starting from the hypothesis for S_0 . Given S_i with $|S_i| \geq 2$, split it into two halves. The ball of S_i is the union of the balls of the halves, so one half S_{i+1}

has $|B_{H'}^{(i+1)k}(S_{i+1})| > |V|/4 \geq u - \frac{1}{2}$. Choose u vertices in this ball. They lie in V , so their own ball of radius k has more than $|V|/2$ vertices. Concatenating paths shows that this ball lies in $B_{H'}^{(i+2)k}(S_{i+1})$. After $j \leq \lceil \log u \rceil \leq L + 1$ steps we reach a single vertex $x \in S_0$ with $|B_{H'}^{(j+1)k}(\{x\})| > |V|/2$, and $(j+1)k \leq 2kL$. So x is good, which is a contradiction. $\square$

Claim 4.2.2. *Suppose that $|V| \geq \rho N/2$ and that $|B_{H-F}^k(U)| > |V|/2$ for all nonempty $U \subseteq V(H)$ and all $F \subseteq E(H)$ with $|F| \leq \bar{s}|U|$. Then H links through V with multiplicity t .*

Proof. Let $\pi_1, \dots, \pi_q$ be a multiset of pairs in which every vertex lies in at most t pairs. Let $\mathcal{A}_i$ be the family of edge sets of paths through V of length at most h joining the pair π_i . We check the condition of [Theorem 2.4](#). Let $I \subseteq [q]$ be nonempty and let S be a set of edges with $|S| \leq (2h-1)(|I|-1)$. Choose a maximal $I' \subseteq I$ whose pairs are pairwise disjoint, and let $q' = |I'|$. Every pair of I meets one of the $2q'$ vertices covered by I' , and each of these lies in at most t pairs, so $|I| \leq 2tq'$. Let $u = \lceil \rho q'/10 \rceil$. Then $2u-1 \leq q'$, and $4u-2 \leq \rho N/5 + 2 \leq |V|$, because $q' \leq N/2$. Moreover

$$|S| < 2h|I| \leq 4htq' \leq \frac{40ht}{\rho}u = \bar{s}u.$$

So every set U of u vertices has $|B_{H-S}^k(U)| > |V|/2$. Apply [Claim 4.2.1](#) to $H' = H - S$ and $2u-1$ of the pairs in I' . It gives a path of length at most $4kL \leq h$ in $H - S$ through V that joins some pair π_i with $i \in I$. Its edge set lies in $\mathcal{A}_i$ and avoids S . By [Theorem 2.4](#) we can choose pairwise disjoint members of the $\mathcal{A}_i$, that is, edge-disjoint paths. $\square$

Large balls. In [Claims 4.2.3–4.2.7](#), H' is an s' -expander on $V(H)$ with $s' \geq \theta + 1$. The next claim replaces the case analysis of [\[4, Proposition 13\]](#).

Claim 4.2.3 (A growth certificate). *Let $1 \leq |W| \leq 2N/3$, $F \subseteq E(H')$ with $|F| \leq s'|W|/4$, and let d be an integer with $1 \leq d \leq s'$. Let Γ be a maximal set of edges of $H' - F$ between W and $V(H) \setminus W$ in which every vertex has degree at most d . Then $|\Gamma| \geq \alpha d|W|/4$.*

Proof. Suppose not. Then the sets $S \subseteq W$ and $X \subseteq V(H) \setminus W$ of vertices of Γ -degree exactly d each have fewer than $\alpha|W|/4$ elements. Let $U = W \setminus S$. By maximality, every edge of $H' - F - \Gamma$ from U to $V(H) \setminus W$ ends in X . So $N_{H'-F-\Gamma}(U) \subseteq S \cup X$, a set of fewer than $\alpha|W|/2$ vertices. But $|U| \geq 3|W|/4$ and $|F \cup \Gamma| \leq s'|W|/4 + \alpha d|W|/4 \leq s'|W|/2 \leq s'|U|$. So expansion gives $|N_{H'-F-\Gamma}(U)| \geq \alpha|U| \geq 3\alpha|W|/4$, which is a contradiction. $\square$

Claim 4.2.4 (One random batch). *In the setting of [Claim 4.2.3](#) with $d = \lceil 1/p \rceil$, let $S \subseteq W$ contain each vertex independently with probability p . Then, with probability at least $1 - \exp(-\alpha p|W|/300)$, at least $\alpha|W|/16$ vertices outside W have an $(H' - F)$ -neighbour in S .*

Proof. Let Z be the number of vertices outside W with a Γ -neighbour in S . A vertex of Γ -degree $r \geq 1$ is counted with probability $1 - (1-p)^r \geq pr/2$, since $pr \leq pd \leq 1 + p \leq 3/2$. So $\mathbb{E}Z \geq p|\Gamma|/2 \geq \alpha|W|/8$. Changing whether $w \in W$ lies in S changes Z by at most $\deg_\Gamma(w) \leq d$, and $\beta = \sum_w p(1-p) \deg_\Gamma(w)^2 \leq pd|\Gamma| \leq 2d\mathbb{E}Z$. By [Lemma 2.6](#) with $\lambda = \mathbb{E}Z/2$ and $d \leq 2/p$,

$$\mathbb{P}(Z < \tfrac{1}{2}\mathbb{E}Z) \leq \exp\left(-\frac{3\mathbb{E}Z}{52d}\right) \leq \exp\left(-\frac{\alpha p|W|}{300}\right). \quad \square$$

Call a set A *well-expanding* if $|N_{H'}(A)| \geq \theta|A|$.

Claim 4.2.5 (Sprinkling). *Let A be well-expanding with $a = |A| \geq 1$, and let $F \subseteq E(H')$ with $|F| \leq a$. Then $\mathbb{P}(|B_{H'-F}^k(A)| \leq 0.56\rho N) \leq N^{-10a}$.*

Proof. Realise V as $V_1 \cup \dots \cup V_k$ with independent random sets: V_i is p -random for $i < k$, and V_k is q -random with $(1-p)^{k-1}(1-q) = 1-\rho$. As $(k-1)p \leq \rho/10$, we have $q \geq 0.9\rho$. For $1 \leq i \leq k$ let B_i be the set of vertices reachable from A by a path in $H' - F$ of length at most i through $V_1 \cup \dots \cup V_{i-1}$. Then $B_1 = A \cup N_{H'-F}(A)$ has at least θa vertices. If a vertex outside B_i has an $(H' - F)$ -neighbour in $B_i \cap V_i$, then it lies in B_{i+1} .

Condition on $V_1, \dots, V_{i-1}$ and suppose $|B_i| \leq 2N/3$. [Claim 4.2.4](#) applies with $W = B_i$: we have $|F| \leq a \leq s'|B_i|/4$ and $d = \lceil 1/p \rceil \leq \theta + 1 \leq s'$. So $|B_{i+1}| \geq (1 + \alpha/16)|B_i|$, except with probability at most $\exp(-\alpha p \theta a/300) = \exp(-C_1 L a/300) \leq N^{-12a}$. Since $(1 + \alpha/16)^{k-1} > N$, with probability at least $1 - kN^{-12a}$ we reach $|B_k| > 2N/3$. Every vertex of $B_k \cap V_k$ lies in $B_{H'-F}^k(A)$. Given $|B_k| > 2N/3$, the size of $B_k \cap V_k$ is binomial with mean at least $0.6\rho N$. So by [Lemma 2.5](#) it is at most $0.56\rho N$ with probability at most $e^{-\rho N/750} \leq e^{-\rho \theta a/750} \leq N^{-11a}$. Here we used $N \geq \theta a$ and $\rho \theta \geq 10^4 C_1 L$. $\square$

Claim 4.2.6 (Well-expanding subsets). *Every U with $1 \leq |U| \leq 2N/3$ contains a well-expanding set A with $|A| \geq \alpha|U|/(3\theta) = 2\mu_0|U|$.*

Proof. This is [[4](#), Proposition 18]. Let $A \subseteq U$ be a maximal well-expanding set ($\emptyset$ is well-expanding). If $A = U$ we are done. Otherwise, by maximality every $v \in U \setminus A$ has fewer than $\theta + 1$ neighbours outside $A \cup N_{H'}(A)$; adding v to A would otherwise create a larger well-expanding set. Delete these fewer than $(\theta + 1)|U| \leq s'|U|$ edges. Afterwards the external neighbourhood of U lies in $N_{H'}(A)$, so expansion gives $\alpha|U| \leq |N_{H'}(A)|$. In particular $A \neq \emptyset$. On the other hand, $A \cup \{v\}$ is not well-expanding, so $|N_{H'}(A)| < \theta(|A| + 1) + 1 \leq 3\theta|A|$. $\square$

Claim 4.2.7. *With probability at least $1 - 2N^{-6}$, $|B_{H'-F}^k(U)| > 0.56\rho N$ for every nonempty U and every $F \subseteq E(H')$ with $|F| \leq \mu_0|U|$.*

Proof. Take a union bound of [Claim 4.2.5](#) over all pairs (A, F) with A well-expanding, $|A| = a$ and $|F| \leq a$. There are at most N^{3a+1} such pairs for each a , so the failure probability is at most $\sum_a N^{1-7a} \leq 2N^{-6}$. If $|U| \leq 2N/3$, [Claim 4.2.6](#) gives a well-expanding $A \subseteq U$ with $|A| \geq 2\mu_0|U| \geq |F|$, and $B_{H'-F}^k(U) \supseteq B_{H'-F}^k(A)$. If $|U| > 2N/3$, apply this to a subset $U' \subseteq U$ of size $\lceil N/2 \rceil$, for which $|F| \leq \mu_0 N \leq 2\mu_0|U'|$. $\square$

Amplification. Colour the edges of H uniformly at random with K colours, independently of V , and let $H_1, \dots, H_K$ be the colour classes. Consider the events

$$(E1) \quad \rho N/2 \leq |V| \leq 1.1\rho N;$$

$$(E2) \quad \text{every } H_i \text{ is an } (s/2K)\text{-expander, where } s/2K \geq \theta + 1;$$

$$(E3) \quad \text{the conclusion of Claim 4.2.7 holds for every } H' = H_i.$$

Note that $\rho N \geq (C_* L^{18} N^2)^{1/3}$, because $\rho^3 \geq C_* L^{18}/s$ and $s < N$. By [Lemma 2.5](#), [Lemma 4.1](#) with $p = 1/K$, and [Claim 4.2.7](#), all three events hold with probability at least $1 - 2e^{-\rho N/300} - KN^{-10} - 2KN^{-6} \geq 1 - N^{-3}$, since $K < s < N$. Assume they hold. Let U be nonempty and $|F| \leq \bar{s}|U|$. Some colour i has $|F \cap E(H_i)| \leq \bar{s}|U|/K \leq \mu_0|U|$, so

$$|B_{H-F}^k(U)| \geq |B_{H_i-F}^k(U)| > 0.56\rho N \geq |V|/2.$$

By [Claim 4.2.2](#), H links through V with multiplicity t . The colouring was only an auxiliary device. Linking is a property of V alone, and it fails with probability at most N^{-3} . $\square$

5 Three decomposition lemmas

This section proves the three local moves of the proof. They are stated without reference to the hierarchy. In [Section 6](#) they are applied once per piece ([Lemma 5.1](#)) and once per round ([Lemmas 5.2](#) and [5.3](#)).

5.1 Clearing a set of vertices

Lemma 5.1 (Clearing). *There is an absolute constant N_0 such that the following holds. Let O be an s -expander with vertex set Z , where $N = |Z| \geq N_0$ and $s \geq \log^{40} N$. Let $E \supseteq E(O)$ be a set of edges with both ends in Z , and let $P \subseteq Z$. Then there is a set $O' \subseteq E(O)$ with*

$$f(E[P] \cup O') \leq 101|P|.$$

So all edges inside P can be removed at a constant cost per vertex of P , using some further edges of O . Every edge of $E \setminus (E[P] \cup O')$ has an end outside P .

The proof uses a *vortex*, a nested sequence of random sets $P = P_0 \supseteq P_1 \supseteq \dots \supseteq P_J$ whose sizes halve. This device comes from iterative absorption [2]. At step j we decompose all remaining edges that touch the layer $W_j = P_j \setminus P_{j+1}$ into paths (Lemma 2.2). We extend each path at both ends so that it ends in the next level, and close it into a cycle by a connecting path through a random set inside the next level (Figure 5.1). Closing paths through path-connecting random sets is the method of [4]. Combining it with a vortex is due to Coffey [5]. What survives all J steps lives on the tiny set P_J and is decomposed by Lemma 2.3.

Proof. Let $L = \log N$, and assume $P \neq \emptyset$.

Outlets. Let $m = \lceil L^6 \rceil$ and $b = \lceil 64mL^2 \rceil = \lceil 2mL^2/\varepsilon \rceil$. We claim that every vertex w has a set $X(w)$ of m neighbours in O , its *outlets*, such that every vertex is an outlet of at most b vertices. Form a bipartite graph. On the left are the ordered pairs (w, y) with $wy \in E(O)$. On the right, every vertex y has b *load* slots and every vertex w has $\deg_O(w) - m \geq 0$ *spare* slots. Join (w, y) to the load slots of y and to the spare slots of w . A matching that covers the left side gives, for every w , at least m pairs (w, y) matched to load slots, and we let $X(w)$ consist of m of these vertices y .

We check Hall's condition [9]. Let Λ be a set of pairs, let R be the set of their second coordinates, and let S' be the set of w that occur in more than $\deg_O(w) - m$ pairs of Λ . Then $|\Lambda| \leq (\text{spare slots}) + m|S'|$, so it suffices that $b|R| \geq m|S'|$. This holds trivially if $S' = \emptyset$. Otherwise every $w \in S'$ has fewer than m neighbours outside R . Take $U \subseteq S'$ with $|S'|/2 \leq |U| \leq 2N/3$, and delete the at most $(m-1)|U| \leq s|U|$ edges from U to outside R . Then the external neighbourhood of U lies in R , so $|R| \geq \varepsilon|U|/L^2 \geq \varepsilon|S'|/(2L^2)$, and $b|R| \geq m|S'|$.

The random vortex. Let $J = \lfloor \log(L/6) \rfloor$, so that $6/L \leq 2^{-J} < 12/L$. Make the following independent random choices.

- Colour every edge of O uniformly with one of $2J+1$ colours: a *reserve* colour and *connector* colours (j, c) , for $0 \leq j < J$ and $c \in \{1, 2\}$. Let O^{res} and $O_{j,c}$ be the colour classes.
- Give every $v \in P$ a level $\text{lev}(v) \in \{0, \dots, J\}$ with $\mathbb{P}(\text{lev}(v) \geq j) = 2^{-j}$, and put $\text{lev}(v) = J$ for $v \notin P$. Let $U_j = \{v : \text{lev}(v) \geq j\}$, $P_j = P \cap U_j$ and $W_j = P_j \setminus P_{j+1}$.
- Give every vertex a label in $\{0, 1, 2\}$, with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$. Let Z_j and $V_{j,c}$ be the sets of vertices of U_{j+1} with labels 0 and c , respectively.

We claim that with positive probability the following hold:

- (V1) every $O_{j,c}$ links through $V_{j,c}$ with multiplicity $b+2$;
- (V2) for every $w \in P$ and $j < J$, at least four edges of O^{res} join w to $X(w) \cap Z_j$;
- (V3) $\sum_{j < J} |P_j| \leq 8|P|$ and $|P_J| \leq 48|P|/L$.

By Lemma 4.1, each $O_{j,c}$ is an $s/(4J+2)$ -expander with probability at least $1 - N^{-10}$. Every vertex lies in $V_{j,c}$ independently with probability at least $2^{-J}/4 \geq 1/L$. So Lemma 4.2, with $t = b + 2 = O(L^8)$ and $\rho = 1/L$, gives (V1) with probability $1 - O(JN^{-3})$. It needs $s/(4J+2) \geq C_*(b+2)L^{21}$, which holds as $s \geq L^{40}$. For (V2), the number of such edges is a sum of independent indicators, one for each $y \in X(w)$, with mean at least $m \cdot \frac{1}{2J+1} \cdot \frac{6}{L} \cdot \frac{1}{2} \geq L^4$. So by Lemma 2.5 it is less than four with probability at most $e^{-L^4/3}$, and a union bound over the NJ pairs (w, j) costs $o(1)$. For (V3), the means are $\sum_j 2^{-j}|P| < 2|P|$ and $2^{-J}|P| < 12|P|/L$, so by Markov's inequality each fails with probability less than $1/4$. Fix an outcome satisfying (V1)–(V3).

Step j , for $j = 0, 1, \dots, J-1$. Let E_j be the set of edges of $E[P]$ not used so far. We maintain that $E_j \subseteq E[P_j]$, which holds for $j = 0$. Let F_j be the set of edges of E_j that meet W_j .

- (1) For every $w \in W_j$ use (V2) to *reserve* four edges of O^{res} from w to distinct outlets in Z_j , two for each *phase* $c \in \{1, 2\}$. Remove the reserved edges from F_j if they lie in it.
- (2) Give every remaining edge of F_j a phase c such that $V_{j,c}$ contains neither of its ends. This is possible: one end lies in W_j , which is disjoint from U_{j+1} , and the other end has only one label.
- (3) For each phase c , decompose the edges of phase c by Lemma 2.2 into at most $|P_j|$ paths, with every vertex an end of at most two of them. At every end $w \in W_j$ of such a path, append a distinct phase- c reserved edge of w . The result is a trail whose ends lie in U_{j+1} , all of whose edges meet W_j , and which avoids $V_{j,c}$. Only its two new end vertices can repeat a vertex. So cutting out closed subtrails splits it into at most two cycles and at most one path P' that joins its two ends, if they differ. All cycles have length at least three, since the graph is simple. The path P' has length at least two, because each of its edges meets W_j and its ends do not.
- (4) For each phase c , join the two ends of every residual path P' by a path through $V_{j,c}$ in $O_{j,c}$, edge-disjointly, using (V1). A vertex of U_{j+1} is an end of at most 2 phase- c paths from the decomposition, and it is the far end of at most one phase- c reserved edge of each w that has it as an outlet. So it lies in at most $b + 2$ pairs. Each connector meets its residual path only at the ends, since its interior lies in $V_{j,c}$, which the residual path avoids. As the residual path has length at least two, the two together form a cycle even if the connector is a single edge.
- (5) Pay for the unused reserved edges.

The edges of E_j that do not meet W_j lie in $E[P_{j+1}]$. So after we also remove the connector edges that lie in $E[P]$, the invariant $E_{j+1} \subseteq E[P_{j+1}]$ holds.

No edge is used twice. Step j uses edges of $F_j \subseteq E_j$, reserved edges wy with $w \in W_j$ and $y \in U_{j+1}$, and connector edges of colours $(j, 1), (j, 2)$. Every connector edge has both ends in U_{j+1} , because its interior lies in $V_{j,c} \subseteq U_{j+1}$ and so do the ends of residual paths. An earlier step $i < j$ uses only edges that meet W_i , reserved edges at W_i , and connector edges of colours $(i, \cdot)$. None of these is a reserved or connector edge of step j , because W_i is disjoint from $U_{i+1} \supseteq U_j$. Within step j , the connectors of one phase are edge-disjoint, the two phases use different colours, and connector edges do not meet W_j , so they are not in F_j .

The finish and the count. After step $J-1$ the unused edges of $E[P]$ lie in $E[P_J]$. By Lemma 2.3 and (V3) they cost at most $|P_J|(\log |P_J| + 1) \leq |P_J|(L + 1) \leq 49|P|$ objects. Step j produces at most three cycles per path and pays for at most $4|W_j|$ reserved edges. So all steps together cost at most $6 \sum_j |P_j| + 4|P| \leq 52|P|$ objects. The total is at most $101|P|$, and O' consists of the reserved and connector edges that were used. $\square$

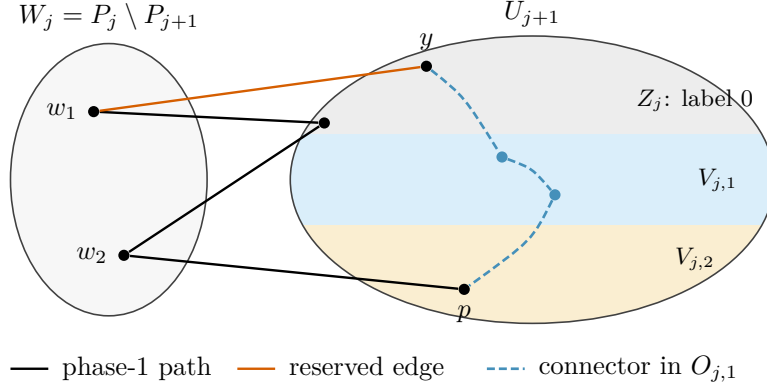


Figure 5.1: One step of the clearing lemma. A phase-1 path of remaining edges at W_j avoids $V_{j,1}$. Its end $w_1 \in W_j$ is extended by a reserved edge to an outlet $y \in Z_j$. Its other end p already lies in P_{j+1} . The two ends, both in U_{j+1} , are joined by a connector through $V_{j,1}$, which closes a cycle. Edges from P_{j+1} to later levels are handled in later steps.

5.2 Closing cherries

The second lemma chains cherries into cycles. In the application the ports are the ports of one round, the classes are given by their anchors, and the reservoirs are edges lent by the anchors.

Lemma 5.2 (Cherries). *Let B be a set of edges and let Π be a set of vertices, called ports, partitioned into classes Π_Y indexed by a family $\mathcal{Y}$ of vertex sets with $\Pi_Y \subseteq Y$. Let $d, K \geq 1$ be integers. Suppose that:*

- (C1) *every edge of B has an end in Π , and an edge of B at a port of Π_Y has its other end outside Y ;*
- (C2) *every port is incident with at most d edges of B ;*
- (C3) *for each $Y \in \mathcal{Y}$ there are pairwise disjoint sets $T_{Y,1}, \dots, T_{Y,K} \subseteq Y \setminus \Pi_Y$ and graphs $R_{Y,1}, \dots, R_{Y,K}$ such that $R_{Y,j}$ links through $T_{Y,j}$ with multiplicity d ;*
- (C4) *the edge sets of all graphs $R_{Y,j}$ are pairwise disjoint and disjoint from B .*

For $Y \in \mathcal{Y}$ let $\kappa_Y = \max_x |\{u \in \Pi_Y : xu \in B\}|$. Then there are $J \subseteq B$ and a set R' of edges of the graphs $R_{Y,j}$ such that $(B \setminus J) \cup R'$ decomposes into at most

$$\sum_{Y: \Pi_Y \neq \emptyset} \left(\frac{\kappa_Y}{2} + 2d \right) + \frac{|B|}{2K}$$

cycles, and for every $x \notin \Pi$ and every Y , at most one edge of J joins x to Π_Y .

Proof. Assign every edge of B to one of its ends in Π ; edges at non-ports are assigned to their port end. For an edge assigned to $u \in \Pi_Y$, call its other end, which lies outside Y , its *centre*, and call Y its class. For each class Y and centre x , pair up the edges of class Y with centre x into *cherries* uxv with $u, v \in \Pi_Y$. If their number is odd, put one of them into J . Every edge at a non-port x has centre x , so J has the stated property.

Fix a class Y with N_Y cherries. Two cherries *conflict* if they share a vertex. A cherry uxv conflicts with at most $d - 1$ others at each of u and v . Ends lie in Y and centres do not, so a cherry can share u or v only as an end. At the centre, a cherry conflicts with at most $\kappa_Y/2 - 1$ others. So the conflict degree is at most $\kappa_Y/2 + 2d - 3$. Put the cherries greedily into groups of at most K pairwise vertex-disjoint cherries, opening a new group only

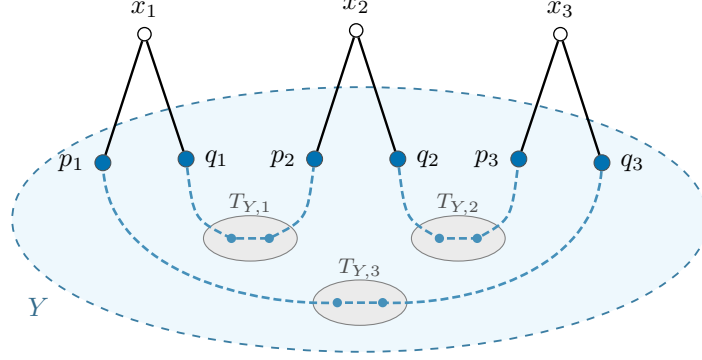


Figure 5.2: Closing a group of three vertex-disjoint cherries $p_i x_i q_i$ whose ends lie in the anchor Y and whose centres lie outside it. The connector from q_i to p_{i+1} runs through its own slot $T_{Y,i}$, so the connectors are internally disjoint from each other and from the cherries, and the result is one cycle.

if every existing group is full or contains a conflicting cherry. At most $\kappa_Y/2 + 2d - 3$ groups contain a conflicting cherry, and fewer than N_Y/K groups are full. So there are at most $\kappa_Y/2 + 2d - 3 + \lceil N_Y/K \rceil$ groups.

Order each group as $p_1 x_1 q_1, \dots, p_k x_k q_k$, with $k \leq K$, and join q_i to p_{i+1} (indices modulo k) by a path through $T_{Y,i}$ in $R_{Y,i}$ (Figure 5.2). For each slot i , we request these connections for all groups of class Y at once. Each port is an end of at most d cherries, so it lies in at most d requests for slot i , and (C3) provides edge-disjoint connectors. In a group, the connectors have their interiors in the pairwise disjoint sets $T_{Y,i} \subseteq Y \setminus \Pi_Y$. These sets avoid the ends of the cherries, which are ports of Π_Y , and the centres, which lie outside Y . So each group becomes a cycle; when $k = 1$ the cycle is $p_1 x_1 q_1$ followed by a connector from q_1 back to p_1 . Different groups and different classes use disjoint edges by (C4). Since $\sum_Y N_Y \leq |B|/2$, summing the number of groups gives the bound. $\square$

5.3 The quotient

The third lemma turns the leftover edges into edges of an auxiliary graph whose cycles lift to cycles of G . Leftovers are of two kinds: *hub edges* au from a hub a to a port u , and *links* uv or uxv between two ports, possibly through a middle vertex x . Every end at a port is attached to a pool vertex by a private *donor edge* (Figure 5.3). Two things must be controlled. First, the auxiliary graph must be simple. This is achieved by splitting parallel edges into layers. Second, it must be small. Hubs are grouped so that each hub has few copies, and the random choices of donors spread the links evenly over the pools.

Lemma 5.3 (Quotient). *Let $d \geq 2$, $g \geq 4d + 4$ and $n \geq 1$ be integers. Let A , U , I , W^- and W^+ be pairwise disjoint sets of vertices of a graph Γ (hubs, ports, middles and two pools), and let $W = W^- \cup W^+$. Suppose we are given:*

- *a set $\mathcal{H}$ of hub edges au with $a \in A$ and $u \in U$, and a set $\mathcal{L}$ of links, each an edge uv or a path uxv of Γ with $u \neq v \in U$ and $x \in I$, all edges involved being distinct;*
- *every port lies on at most d hub edges and links, every middle lies on at most d links, and $|\mathcal{H}|, |\mathcal{L}| \leq nd$;*
- *for every port u , donor sets $D^-(u) \subseteq N_\Gamma(u) \cap W^-$ and $D^+(u) \subseteq N_\Gamma(u) \cap W^+$, each of size at least g ;*

- a function $\beta : W \rightarrow \mathbb{Z}_{\geq 0}$ such that for all $a \in A$ and $w \in W$,

$$|\{u : au \in \mathcal{H}, w \in D^-(u) \cup D^+(u)\}| \leq \beta(w). \quad (5.1)$$

Then there are a set $\mathcal{H}_{\text{paid}} \subseteq \mathcal{H}$ and a simple graph Q with

$$|\mathcal{H}_{\text{paid}}| + |V(Q)| \leq |A| + d^3 \sum_{w \in W} \beta(w) + 3d[2 \log d]|W| + 30n\left(\frac{d}{g} + \frac{1}{d}\right),$$

such that whenever Q has a decomposition into k objects, the set $(\mathcal{H} \setminus \mathcal{H}_{\text{paid}}) \cup E(\mathcal{L}) \cup D'$ has a decomposition into at most $2k$ objects for some set D' of donor edges uw with $w \in D^\pm(u)$.

Donor edges are never hub edges or edges of links, because W is disjoint from $A \cup U \cup I$.

Proof. Layers. Let $\mathcal{M}$ be a loopless multigraph, and let $m(v, w)$ be the number of edges between v and w . Its *layers* are the simple graphs $\mathcal{M}_1, \mathcal{M}_2, \dots$, where $\mathcal{M}_i$ contains the i -th edge of each parallel class, and isolated vertices are removed. Their total number of vertices is $\sum_v \max_w m(v, w)$.

Hub edges. For every hub a , split the hub edges at a into groups of size $\lfloor g/2 \rfloor$, except for at most one smaller group. There are at most $|A| + 4|\mathcal{H}|/g$ groups. Colour the groups independently and uniformly with d^3 colours, and give each hub edge the colour of its group. Let $\mathcal{H}_{\text{paid}}$ be the set of hub edges au for which another hub edge at u has the same colour. Two hub edges at u come from different hubs, hence from different groups, so $\mathbb{P}(au \in \mathcal{H}_{\text{paid}}) \leq (d-1)/d^3$ and $\mathbb{E}|\mathcal{H}_{\text{paid}}| \leq |\mathcal{H}|/d^2 \leq n/d$. For each hub edge $au \notin \mathcal{H}_{\text{paid}}$ choose a donor $w(au) \in D^-(u) \cup D^+(u)$. The donors must be distinct for distinct hub edges at the same port and for hub edges in the same group. This can be done greedily, since fewer than $d + g/2 \leq g$ choices are excluded. For each colour γ , let $\mathcal{M}_\gamma$ be the multigraph with an edge $aw(au)$ for every hub edge $au \notin \mathcal{H}_{\text{paid}}$ of colour γ . A group contributes at most one edge to any pair (a, w) , so the copies of a in the layers of all $\mathcal{M}_\gamma$ number at most the number of groups at a . Distinct hub edges at a with the same donor w have distinct ports, so by (5.1) every pool vertex w has at most $\beta(w)$ copies per colour. These layers therefore have at most $|A| + 4nd/g + d^3 \sum_w \beta(w)$ vertices.

Links. Colour the links greedily with at most $3d$ colours so that links of the same colour share no port and no middle; a link meets at most $3(d-1)$ others. Orient every link from one end port to the other. For every port u , choose independent uniformly random injections. One maps the links leaving u into $D^-(u)$, and the other maps the links entering u into $D^+(u)$; both avoid the donors of hub edges at u . Their domains have size at most d , and their ranges have size at least $g - d \geq g/2$. A link from u to v thus gets a *tail junction* $w^- \in W^-$ and a *head junction* $w^+ \in W^+$. For each colour γ , let $\mathcal{N}_\gamma$ be the multigraph with an edge w^-w^+ for every link of colour γ , and split it into layers.

To bound the number of vertices we use an occupancy estimate. Let $\xi_1, \dots, \xi_q$ be independent random variables, each taking every value with probability at most π , and let $X_y = |\{i : \xi_i = y\}|$. Then for every integer $t \geq 1$

$$\mathbb{E} \max_y X_y \leq t + 2e\pi q + 4e2^{-t}q. \quad (5.2)$$

Indeed, $\mathbb{P}(X_y \geq j) \leq \mathbb{E} \binom{X_y}{j} \leq (\mathbb{E} X_y)^j / j! \leq (e\mathbb{E} X_y / j)^j$. For $j \geq T = \max(t, \lceil 2e\pi q \rceil)$ we have $e\mathbb{E} X_y / j \leq 1/2$, so this is at most $(e\mathbb{E} X_y / j)2^{1-j}$, and summing gives $\mathbb{E}[X_y \mathbf{1}_{X_y \geq T}] \leq 4e2^{-T} \mathbb{E} X_y$. Now use $\max_y X_y \leq T - 1 + \sum_y X_y \mathbf{1}_{X_y \geq T}$ and $\sum_y \mathbb{E} X_y = q$.

Fix a colour γ and condition on all tail junctions. For a pool vertex $w \in W^-$, the links of colour γ with tail junction w have distinct head ports. So their head junctions

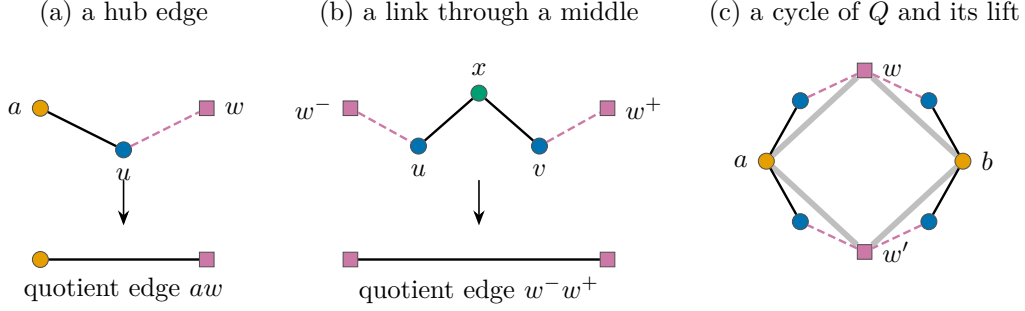


Figure 5.3: The quotient. (a) A hub edge au , extended by a donor edge uw (dashed) to a pool vertex, becomes the edge aw of Q . (b) A link uxv between two ports becomes an edge between two pool vertices. (c) The cycle $awbw'$ of Q (grey) lifts to an 8-cycle of G . The branch vertices are hubs and pool vertices, and the private ports in between keep the lift simple.

are independent, each uniform on a set of size at least $g/2$. The number of copies of w in the layers of $\mathcal{N}_\gamma$ is the maximum multiplicity of an edge at w . By (5.2) with $t = \lceil 2 \log d \rceil$ and $\pi = 2/g$, its expectation is at most $t + 4eq_w/g + 4eq_w/d^2$, where q_w is the number of these links. The same holds for W^+ , with tails and heads exchanged. Summing over w and over the $3d$ colours, the expected number of vertices in the layers of all $\mathcal{N}_\gamma$ is at most $3dt|W| + 8end/g + 8en/d$.

The graph Q . Let Q be the vertex-disjoint union of all layers of all $\mathcal{M}_\gamma$ and $\mathcal{N}_\gamma$. Adding up, $\mathbb{E}(|\mathcal{H}_{\text{paid}}| + |V(Q)|)$ is at most the bound in the lemma, since $4 + 8e < 30$ and $1 + 8e < 30$. Fix an outcome of the random choices that achieves this bound.

Lifting. An edge aw of Q that comes from the hub edge au represents the path auw , and an edge w^-w^+ that comes from a link represents the path w^-uvw^+ or w^-uxvw^+ . These paths have three properties.

- (L1) Distinct edges of Q represent edge-disjoint paths. The cores (hub edges and links) are distinct, and the donor edges at a port are distinct by construction.
- (L2) In each layer, distinct vertices are distinct vertices of Γ . These are hubs and pool vertices.
- (L3) In each layer, the interior vertices of the represented paths are pairwise distinct and are not vertices of the layer. In one colour a port lies on at most one surviving hub edge, links of one colour share no port or middle, and ports and middles are neither hubs nor pool vertices.

Take a decomposition of Q into k objects. A cycle of Q lies in one layer, and by (L2) and (L3) replacing each of its edges by the represented path gives a cycle of Γ . For a single edge of Q we output only its core: the hub edge, or the one or two edges of the link. Its donor edges are left unused. By (L1) this gives at most $2k$ edge-disjoint objects covering exactly $(\mathcal{H} \setminus \mathcal{H}_{\text{paid}}) \cup E(\mathcal{L}) \cup D'$, where D' is the set of donor edges on lifted cycles. $\square$

6 Proof of the reduction theorem

Let G be a graph on n vertices. If $2e(G)/n < D$ we pay for all edges, and $f(G) < Dn/2$. Otherwise we build the hierarchy of Section 3.3, with rounds $1, \dots, R$, and use the notation of Section 3. This section has four steps. We equip every piece with random reservoirs

(Section 6.1), sacrifice a few ports and choose pools (Section 6.2), decompose the rounds in reverse order (Section 6.3), and count (Section 6.4).

6.1 Reservoirs

Let Y be a piece of round r . It lends to the rounds $\ell \in \{r+2, \dots, R\}$. Put

$$k_r = \sum_{\ell=r+2}^R (M_\ell^2 + 1).$$

By Lemma 3.5 we have $M_\ell \leq M_{r+2} 2^{-(\ell-r-2)}$ and $M_{r+2} \leq \mu_r^{211}$, so $k_r \leq 4\mu_r^{422} \leq \lambda_r^4/2$.

Definition 6.1 (Reservoirs). Colour every edge of H_Y independently at random. It gets the *own* colour with probability $1/2$. Each of the following colours has probability $1/(2k_r)$: for every $\ell \in \{r+2, \dots, R\}$, the *closing* colours (ℓ, j) with $j \in [M_\ell^2]$, and the *donor* colour ℓ . If $r+2 > R$, all edges get the own colour. Let O_Y , $R_{Y,\ell,j}$ and $R_{Y,\ell}^{\text{don}}$ be the spanning subgraphs of H_Y formed by these colour classes. Independently, for every ℓ , give every vertex of Y a *slot* for round ℓ . It is slot j with probability M_ℓ^{-4} for each $j \in [M_\ell^2]$, and no slot otherwise. Let $T_{Y,\ell,j}$ be the set of vertices with slot j for round ℓ . The piece Y is *good* if

(G1) O_Y is an $(s_r/4)$ -expander, and

(G2) every $R_{Y,\ell,j}$ links through $T_{Y,\ell,j}$ with multiplicity M_ℓ .

Otherwise it is *bad*.

The closing colours of round ℓ close cherries in round ℓ (Lemma 5.2), and the donor colour supplies donor edges for the quotient of round ℓ (Lemma 5.3).

Lemma 6.2. *Every piece Y is bad with probability at most $|Y|^{-2}$.*

Proof. Let $N = |Y|$, so $\lambda_r^{103} \leq N \leq M_r$ and $\log N \leq 3\lambda_r$. By Lemma 4.1 with $p = 1/2$, (G1) fails with probability at most N^{-10} . Each $R_{Y,\ell,j}$ is a random subgraph with $p = 1/(2k_r) \geq \lambda_r^{-4}$. So by Lemma 4.1 it is a $(\lambda_r^{96}/2)$ -expander with probability at least $1 - N^{-10}$. The set $T_{Y,\ell,j}$ is M_ℓ^{-4} -random and independent of the colouring. Lemma 4.2 with $t = M_\ell$ and $\rho = M_\ell^{-4}$ needs $\lambda_r^{96}/2 \geq C_* M_\ell^{13} (3\lambda_r)^{18}$, which holds since $M_\ell \leq \mu_r^{211}$. So (G2) fails for a given (ℓ, j) with probability at most $N^{-10} + N^{-3}$. There are at most $k_r \leq \lambda_r^4 \leq N^{1/20}$ pairs (ℓ, j) , so the total failure probability is at most N^{-2} . $\square$

6.2 Lost ports, pools and donors

Definition 6.3 (Lost and live ports). A port u of round ℓ with anchor $Y = \text{anc}_\ell(u)$ is *lost* if Y is bad or u lies in some slot set $T_{Y,\ell,j}$. Otherwise it is *live*. Let Lost_ℓ be the set of lost ports of round ℓ .

So a live port avoids all slot sets of its anchor for its round. That is what Lemma 5.2 requires. If a piece is bad, every port anchored at it in every round is lost, so a bad piece never lends anything.

Pools. Independently of all previous choices, give every vertex of G at most one *pool label* (ℓ, r, σ) , with $3 \leq \ell \leq R$, $1 \leq r \leq \ell - 2$ and $\sigma \in \{-, +\}$. Each label has probability $q_\ell 2^{-(\ell-r)}$, where $q_\ell = M_\ell^{-5}$. The total probability is at most $\sum_\ell q_\ell < 1$. Let $W_{\ell,r,\sigma}$ be the set of vertices with label (ℓ, r, σ) , let $W_\ell^\sigma = \bigcup_r W_{\ell,r,\sigma}$ and $W_\ell = W_\ell^- \cup W_\ell^+$, and for $w \in W_\ell$ let $r(w)$ be the second coordinate of its label. Every vertex lies in W_ℓ with probability less than q_ℓ . The pool of round ℓ is split by the round r of the anchors it serves. This makes condition (5.1) of Lemma 5.3 summable over r (see the proof of Lemma 6.5).

Donors. Let u be a port of round ℓ with anchor Y of round r . Its donor sets are

$$D^\sigma(u) = N_{R_{Y,\ell}^{\text{don}}}(u) \cap W_{\ell,r,\sigma} \quad (\sigma \in \{-, +\}).$$

Let $g_\ell = M_\ell^{10}$. The port u is *deficient* if $|D^-(u)| < g_\ell$ or $|D^+(u)| < g_\ell$. Let Def_ℓ be the set of deficient ports of round ℓ .

Lemma 6.4. *Every port of round ℓ is deficient with probability at most $2e^{-g_\ell}$.*

Proof. For a fixed σ , $|D^\sigma(u)|$ is a sum of independent indicators, one for each of the more than s_r neighbours w of u in H_Y : the edge uw has the donor colour ℓ , and $w \in W_{\ell,r,\sigma}$. By Lemma 3.5(b) its mean is at least

$$\lambda_r^{100} \cdot \lambda_r^{-4} \cdot M_\ell^{-5} 2^{-(\ell-r)} \geq \lambda_r^{95} M_\ell^{-5} \geq 4M_\ell^{10} = 4g_\ell.$$

By Lemma 2.5 it is less than g_ℓ with probability at most e^{-g_ℓ} . $\square$

Table 2 summarises the random choices made so far. They are all independent, and they are made after the hierarchy and the anchors have been fixed.

object	distribution	purpose
own graph $O_Y \subseteq H_Y$	each edge with probability $\frac{1}{2}$	clearing inside Y (Lemma 5.1)
closing graphs $R_{Y,\ell,j}$, $j \in [M_\ell^2]$	each edge with probability $\frac{1}{2k_r}$	closing cherries in round ℓ (Lemma 5.2)
donor graph $R_{Y,\ell}^{\text{don}}$	each edge with probability $\frac{1}{2k_r}$	donor edges in round ℓ (Lemma 5.3)
slot sets $T_{Y,\ell,j} \subseteq Y$	each vertex with probability M_ℓ^{-4}	interiors of cherry connectors
pools $W_{\ell,r,\pm} \subseteq V(G)$	each vertex with probability $q_\ell 2^{-(\ell-r)}$	vertices of the quotient Q_ℓ

Table 2: The random choices for a piece Y of round r and a later round $\ell \geq r + 2$, and for the pools. Lost and deficient ports are the ports for which these choices fail.

The random cost. We now collect, in one random variable, all the costs that depend on these random choices. Let $\mathbf{E}_\ell = \bigcup_{X \in \mathcal{Y}_\ell} E_X$ be the set of edges assigned in round ℓ . For a hub h of round ℓ , let $c_\ell(h)$ be the number of anchors Y such that some edge of $\mathbf{E}_\ell$ joins h to a port with anchor Y . Let

$$\omega_\ell(v) = \begin{cases} c_\ell(v) & \text{if } v \in \text{Hub}_\ell, \\ M_\ell & \text{otherwise.} \end{cases}$$

Let $\beta_r(w)$ be the number of pieces of round r that contain w , and let $t_\ell = \lceil 2 \log M_\ell \rceil$. Define

$$Z = \sum_{\ell=3}^R \left[(101 + M_\ell) |\text{Lost}_\ell| + \sum_{v \in W_\ell} \omega_\ell(v) + M_\ell |\text{Def}_\ell| + M_\ell^3 \sum_{w \in W_\ell} \beta_r(w)(w) + 3M_\ell t_\ell |W_\ell| \right].$$

The weights ω_ℓ are deterministic. They bound the degrees in round ℓ of a leftover graph that will depend on the pools, which lets us control that graph without any independence (see Section 6.3).

Lemma 6.5. $\mathbb{E}Z = o_D(n)$.

Proof. Fix $\ell \geq 3$. A port u with anchor Y of round $r \leq \ell - 2$ has $|Y| \geq m_r \geq M_\ell$, so $\mathbb{P}(u \in \text{Lost}_\ell) \leq |Y|^{-2} + M_\ell^2 \cdot M_\ell^{-4} \leq 2M_\ell^{-2}$ by Lemma 6.2. Hence $\mathbb{E}(101 + M_\ell)|\text{Lost}_\ell| \leq 4n/M_\ell$. Every resident has fewer than M_ℓ edges in $\mathbf{E}_\ell$, because its piece has at most M_ℓ vertices. Every pair (h, Y) counted by $c_\ell(h)$ is witnessed by a distinct edge of $\mathbf{E}_\ell$ at a port, so $\sum_v \omega_\ell(v) \leq 2nM_\ell$. Therefore $\mathbb{E} \sum_{v \in W_\ell} \omega_\ell(v) \leq 2n/M_\ell^4$. By Lemma 6.4, $\mathbb{E} M_\ell |\text{Def}_\ell| \leq 2nM_\ell e^{-g_\ell} \leq n/M_\ell$. Since $\sum_w \beta_r(w) = \sum_{Y \in \mathcal{Y}_r} |Y| < 4n/3$,

$$\mathbb{E} \sum_{w \in W_\ell} \beta_{r(w)}(w) = \sum_{r \leq \ell-2} 2q_\ell 2^{-(\ell-r)} \sum_w \beta_r(w) \leq \frac{4}{3} n q_\ell,$$

so the fourth term has expectation at most $2n/M_\ell^2$. The last term has expectation at most $3M_\ell t_\ell q_\ell n \leq n/M_\ell$. Altogether $\mathbb{E} Z \leq \sum_\ell 10n/M_\ell \leq 20n/D^2$ by Lemma 3.5(a). $\square$

We fix an outcome of all the random choices of Sections 6.1 and 6.2 with $Z \leq \mathbb{E} Z$. From now on everything is deterministic.

6.3 Decomposing the rounds in reverse order

We process the rounds $\ell = R, R-1, \dots, 1$ in this order. We keep a set *Used* of lent edges that rounds already processed have used. It is initially empty.

Round ℓ .

Step 1. *Clear.* For every piece X of round ℓ , let $F_X = E_X \setminus \text{Used}$, let O_X be the own graph of X if X is good and $O_X = H_X$ if X is bad, and let

$$P_X = X \cap (\text{Hub}_\ell \cup \text{Fresh}_\ell \cup \text{Lost}_\ell).$$

Apply Lemma 5.1 to $O = O_X$, $E = F_X$ and $P = P_X$. Let B_X be the set of edges of F_X that remain. Each of them has an end in $X \setminus P_X$, that is, at a live port.

Step 2. *Close cherries* (only if $\ell \geq 3$). Apply Lemma 5.2 to $B = \bigcup_X B_X$. The ports are the live ports of round ℓ , classified by their anchors, and we take $d = M_\ell$, $K = M_\ell^2$, $T_{Y,j} = T_{Y,\ell,j}$ and $R_{Y,j} = R_{Y,\ell,j}$. Add the lent edges used to *Used*. Let J_ℓ be the leftover set.

Step 3. *Pay.* Pay for every edge of J_ℓ with an end in $\text{Lost}_\ell \cup W_\ell \cup \text{Def}_\ell$. Then, at every fresh vertex $x \notin W_\ell$, pair up the remaining edges of J_ℓ at x into links uxv , and pay for the leftover edge if their number is odd.

Step 4. *Quotient.* Apply Lemma 5.3 with $d = M_\ell$, $g = g_\ell$, hubs $A = \text{Hub}_\ell \setminus W_\ell$, ports U the live ports outside $W_\ell \cup \text{Def}_\ell$, middles $I = \text{Fresh}_\ell \setminus W_\ell$, pools $W^\pm = W_\ell^\pm$, and donor sets $D^\pm(u)$. The hub edges are the remaining edges of J_ℓ between A and U . The links are the remaining edges of J_ℓ inside U together with the paths formed in Step 3. We also take $\beta(w) = \beta_{r(w)}(w)$. Pay for the edges of $\mathcal{H}_{\text{paid}}$. Take a decomposition of Q_ℓ into $f(Q_\ell)$ objects, lift it, and add the donor edges used to *Used*.

We now check that every step is legitimate.

Step 1. We show that $O_X \subseteq F_X$, that is, no edge of O_X has been used. Only lent colours are ever used by later rounds. If X is good, O_X consists of own-coloured edges. If X is bad, every port anchored at X is lost. So no cherry of class X is ever closed and no donor set of X is ever used, which means no edge of X is in *Used*. Moreover O_X is an $(s_\ell/4)$ -expander with $s_\ell/4 \geq (3\lambda_\ell)^{40} \geq \log^{40} |X|$ and $|X| \geq m_\ell \geq N_0$. Each resident of X is a hub, a fresh

vertex, a lost port or a live port, so the edges of B_X have an end at a live port. For $\ell \leq 2$ there are no ports, so $P_X = X$ and $B_X = \emptyset$.

Step 2. (C1): Let u be a live port with anchor Y of round r , and let $uy \in B_X \subseteq E(G_\ell)$. By the separation property (H5), $y \notin Y$. (C2): u has fewer than M_ℓ edges in E_X . (C3): The anchor Y of a live port is good, and the sets $T_{Y,\ell,j}$ are pairwise disjoint and avoid the live ports anchored at Y . The graphs $R_{Y,\ell,j}$ link through them by (G2), and their colours are used only in round ℓ , so none of their edges is in Used. (C4): The graphs H_Y are pairwise edge-disjoint, and they are disjoint from $B \subseteq \mathbf{E}_\ell$. The resulting cycles number at most

$$\sum_Y \left(\frac{\kappa_{Y,\ell}}{2} + 2M_\ell \right) + \frac{nM_\ell}{2M_\ell^2},$$

summed over the at most ν_ℓ anchors used in round ℓ . The leftover J_ℓ has at most one edge between each vertex that is not a live port and the live ports of each anchor.

Step 3. All edges of J_ℓ lie in $\mathbf{E}_\ell$. So a non-hub has J_ℓ -degree less than M_ℓ , and a hub h has J_ℓ -degree at most $c_\ell(h)$ by the property of J_ℓ just stated. Hence the edges paid for at pool vertices number at most $\sum_{v \in W_\ell} \omega_\ell(v)$, whatever J_ℓ is. Those paid for at lost and deficient ports number at most $M_\ell(|\text{Lost}_\ell| + |\text{Def}_\ell|)$. At most one edge is paid for at each fresh vertex.

Step 4. After Step 3, every edge of J_ℓ has a live port in U as one end. Its other end is a hub in A , a live port in U , or a fresh vertex in I , since all other cases were paid for in Step 3. So the remaining edges are exactly the hub edges and the edges of links. Ports lie on fewer than M_ℓ of these, middles on fewer than $M_\ell/2$ links, and there are at most nM_ℓ of each kind. The donor sets have size at least $g_\ell \geq 4M_\ell + 4$, and the donor colour ℓ is used only in round ℓ . For condition (5.1): if au is a hub edge and $w \in D^\pm(u)$, then the anchor of u is a piece of round $r(w)$ that contains w . Distinct such ports u have distinct anchors, because J_ℓ has at most one edge from a to the live ports of each anchor. So there are at most $\beta_{r(w)}(w)$ of them.

Every edge is used exactly once. By Lemma 3.4, $E(G)$ is partitioned into the long cycles, the final graph G_{R+1} , whose edges we pay for, and the sets E_X . Fix a piece X of round ℓ . The edges of E_X that are in Used when round ℓ starts were used exactly once, in a later round. A lent colour of X is used only in its target round, and there each edge is used at most once. The remaining edges F_X are used exactly once in round ℓ : by clearing, by a cherry cycle, by a payment, or as part of a hub edge or link in the quotient. The lent edges that round ℓ borrows belong to earlier pieces and are added to Used, so they are not used again when their owners are processed.

6.4 Counting

Table 3 lists all objects produced. Clearing costs at most $101 \sum_\ell \sum_X |P_X|$, and

$$\sum_X |P_X| \leq \sum_X |X \cap \text{Hub}_\ell| + |\text{Fresh}_\ell| + |\text{Lost}_\ell|,$$

since fresh and lost vertices are residents. All other entries follow from Steps 2–4, Lemma 3.7, $Z \leq \mathbb{E}Z = o_D(n)$ and Lemma 3.5(c).

Finally, Lemma 5.3 gives, for each round $\ell \geq 3$,

$$|\mathcal{H}_{\text{paid}}^\ell| + |V(Q_\ell)| \leq |\text{Hub}_\ell| + M_\ell^3 \sum_{w \in W_\ell} \beta_{r(w)}(w) + 3M_\ell t_\ell |W_\ell| + 30n \left(\frac{1}{M_\ell^9} + \frac{1}{M_\ell} \right).$$

Summed over ℓ , the first term is $o_D(n)$ by Lemma 3.7(b), the next two are at most Z , and the last is $o_D(n)$ by Lemma 3.5(c). So $\sum_\ell (|\mathcal{H}_{\text{paid}}^\ell| + |V(Q_\ell)|) = o_D(n)$.

objects	number	reason
long cycles $\mathcal{C}_r$	$o_D(n)$	Lemma 3.7(e)
single edges of G_{R+1}	$< Dn/2$	Lemma 3.4
clearing hubs	$o_D(n)$	Lemma 3.7(b)
clearing fresh vertices	$\leq 202n$	Lemma 3.7(a)
clearing lost ports; paying at lost, pool and deficient vertices	$\leq Z$	Step 3 and Lemma 6.5
cherry cycles	$o_D(n)$	Lemma 3.7(c),(d) and Lemma 3.5(c)
unpaired edges at fresh vertices	$\leq 2n$	Lemma 3.7(a)
paid hub edges in the quotients	$o_D(n)$	Lemma 5.3 , see below
lifted decompositions	$\leq 2 \sum_{\ell} f(Q_{\ell})$	Lemma 5.3

Table 3: The cost of the decomposition. Only fresh vertices, at most twice each, and the final sparse graph cost $\Theta(n)$. Everything charged to rounds is summable.

Proof of [Theorem 1.2](#). Choose D so large that all requirements above hold and that the last sum and all the entries $o_D(n)$ in [Table 3](#) total at most $n/4$. Then $\sum_{\ell} |V(Q_{\ell})| \leq n/4$ and

$$f(G) \leq \left(\frac{D}{2} + 202 + 2 + \frac{1}{4} \right) n + 2 \sum_{\ell} f(Q_{\ell}).$$

So [Theorem 1.2](#) holds with $C_0 = D/2 + 205$. The constant D , and hence C_0 , is absolute, because every requirement on D involves only absolute constants. $\square$

7 Sources

Every ingredient of the proof is listed here with its source.

- *Classical tools.* [Lemma 2.3](#) is the classical $O(n \log n)$ argument. [Lemma 2.2](#) is Lovász’s theorem [17] in the form of [4, Corollary 22]. [Theorem 2.4](#) is Haxell’s theorem [11]. The probabilistic tools are standard [12, 8, 18], and Hall’s theorem [9] is used in [Lemma 5.1](#).
- *Expanders.* Sublinear expanders are due to Komlós and Szemerédi [13, 14], and their robust form to Haslegrave, Kim and Liu [10] and Sudakov and Tomon [20]. [Definition 2.7](#) is [4, Definition 11].
- *Rounds.* The overall scheme of rounds with long-cycle removal, splitting into expanders and polylogarithmic decay of the degree is that of Bucić and Montgomery [4], including the splitting step ([4, Lemma 14]) and the use of long cycles in expanders ([4, Lemma 25]). The depth-first-search proof of [Lemma 3.1](#) is a standard argument (see [15]) and is shorter than the one in [4].
- *Thin cuts.* The rule that moves high-degree vertices into the separator ([Lemma 3.2\(iii\)](#), [Lemma 3.3\(d\)](#)) and its use to bound crowding ([Lemma 3.7\(c\)](#)) are due to Coffey [5].
- *Linking.* [Lemma 4.2](#) extends [4, Theorem 16] to sparse random sets and large multiplicities. Its proof follows [4]: Claims [4.2.1](#) and [4.2.6](#) are [4, Propositions 8 and 18], [Claim 4.2.5](#) follows [4, Lemma 17], and amplification by colouring follows [4, Theorem 16]. The certificate of [Claim 4.2.3](#) and the use of Haxell’s theorem simplify [4, Proposition 13 and Lemma 9]. [Lemma 4.1](#) replaces [4, Lemma 15] and has a standard proof.

- *Anchors, reservoirs, cherries and the quotient.* The idea of lending random colour classes of pieces at least two rounds back, of closing cherries through them, of compressing the leftovers into a quotient graph through private edges to random pools, and of finishing by induction on the number of vertices, is due to Coffey [5]. So is the use of a vortex, in the sense of iterative absorption [2], to clear vertices locally (Lemma 5.1).
- *Simplifications in this exposition.* Compared with [5], the present account uses one kind of piece, kept whole, instead of several kinds of parts; a single clearing lemma; the greedy grouping of Lemma 5.2 instead of a routing scheme for cherry systems; a simplified quotient with a single averaging argument (Lemma 5.3); and the growth certificate in the proof of Lemma 4.2.

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